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Impact of noise channels on hybrid quantum-classical algorithms

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1. Introduction

1.1 Quantum Revolutions

Quantum science is a fundamental branch of physics that describes the behavior of matter and energy at microscopic scales through the principles of quantum mechanics. Since Max Planck introduced the quantum hypothesis in 1900, and the subsequent formulation of matrix mechanics by Werner Heisenberg, Max Born, and Pascual Jordan in 1925, quantum phenomena have been extensively investigated. However, the translation of quantum theory into practical technologies has been gradual and challenging. In general, a physical theory must be thoroughly understood before it can be transformed into reliable and widely accessible devices. The technological developments arising from quantum mechanics are commonly described in terms of two distinct quantum revolutions.

The First Quantum Revolution was based on the discovery and exploitation of quantum properties of matter, such as discrete energy levels, semiconductor behavior, and stimulated emission. It extended from the early studies of energy quantization at the beginning of the twentieth century to the widespread diffusion of quantum-based technologies during the second half of the century. The practical realization of these technologies required several decades of advances in experimental methods, materials science, and engineering.

Once these challenges had been overcome, the first quantum technologies such as the transistor, the laser, and magnetic resonance imaging profoundly transformed modern society. Transistors enabled the miniaturization and large-scale integration of electronic circuits, which underpin modern computing; lasers became essential tools in communications, manufacturing, medicine, and scientific research; and magnetic resonance imaging introduced a powerful non-invasive technique for clinical diagnosis.

The Second Quantum Revolution, which began around the 1980s and continues today, is characterized by the ability to prepare, manipulate, and measure individual quantum systems, such as atoms, electrons, photons, and spins, while preserving their quantum coherence. This level of control makes it possible to exploit fundamental phenomena such as superposition, interference, and entanglement as technological resources.

Within this framework, second-generation quantum technologies are commonly classified into several major fields:

- **Quantum computing and simulation**, which aim to process information and simulate quantum systems by exploiting controllable quantum states;
- **Quantum communication**, which uses quantum states to transmit information and enable secure communication protocols;
- **Quantum sensing, metrology, and imaging**, which exploit quantum effects to improve the sensitivity, precision, or spatial resolution of measurements;
- **Quantum materials**, which investigate and exploit materials whose macroscopic properties arise from quantum phenomena, such as superconductors.

1.2 Quantum Information Theory

The term quantum information theory was first used by Charles H. Bennett around 1970. It refers to the field that combines classical information theory with the principles of quantum mechanics. Quantum information theory explores how quantum systems can be used to process, transmit, and store information in ways that can provide advantages over classical methods. It exploits unique properties of quantum states, such as superposition, entanglement, and interference, to perform certain computational and communication tasks more efficiently than classical approaches.

Its main applications include quantum computing, quantum communication, quantum cryptography, quantum sensing, and quantum error correction. Among these, quantum computing focuses on the use of quantum systems to process information and solve specific computational problems otherwise impossible to solve even for modern supercom-

puters. It is divided into the study of both software, therefore implementation of quantum algorithms, and hardware, which is the hard task to engineer stable and scalable quantum system easy to control. To understand both of them basic knowledge of quantum physics is required, which is mostly linear algebra.

1.3 Quantum Physics Fundamentals

1.3.1 Hilbert space and basis

The state of a quantum system is represented by a vector in a complex Hilbert space, denoted by $\mathcal{H}$. A Hilbert space is a vector space equipped with an inner product, which makes it possible to define the length of a vector and the overlap between two quantum states. Its dimension corresponds to the number of linearly independent basis vectors required to represent any state in the space.

Using the Dirac's bra-ket notation, a state vector is denoted by a ket, $|\psi\rangle$, while its Hermitian conjugate is denoted by a bra, $\langle\psi|$. The inner product between two quantum states $|\phi\rangle$ and $|\psi\rangle$ is written as

$$\langle\phi|\psi\rangle,$$

and it is, in general, a complex number. The simplest non-trivial Hilbert space is the two-dimensional Hilbert Space $\mathbb{C}^2$, that is known as a two-state (or two level) quantum-mechanical system, with its basis given by the two vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The corresponding bra vectors are obtained by taking the Hermitian conjugate of the ket vectors. It follows that

$$\langle 0|1\rangle = \langle 1|0\rangle = 0,$$

while

$$\langle 0|0\rangle = \langle 1|1\rangle = 1.$$

Hence, $|0\rangle$ and $|1\rangle$ are orthonormal and form the computational basis of the two-dimensional Hilbert space $\mathbb{C}^2$. They are the quantum version of a classical bit, which only has the values 0 and 1; the qubit instead is a quantum system that can be prepared in a superposition of two basis states and it's often written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where α and β are complex probability amplitudes satisfying

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1.$$

Although the coefficients α and β are complex numbers, the physical state of a single qubit can be described using only two independent real parameters. This is because an overall global phase does not affect the probabilities of measurement outcomes. Therefore, any pure qubit state can be parametrized as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle,$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$.

This parametrization provides a geometric representation of a single qubit known as the Bloch sphere. Each pure state is associated with a point on the surface of a unit sphere, whose Cartesian coordinates are

$$(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta).$$

The computational basis states are located at the poles of the sphere:

$$|0\rangle \longleftrightarrow (0, 0, 1), \quad |1\rangle \longleftrightarrow (0, 0, -1).$$

The states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

correspond to opposite points on the x -axis. Similarly,

$$|+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad |-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}},$$

correspond to opposite points on the y -axis.

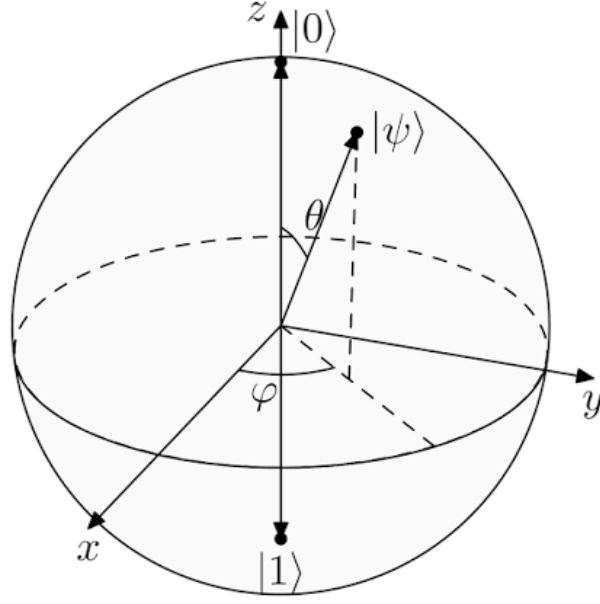


Figure 1.1: Bloch sphere representation of a single-qubit pure state. The computational basis states $|0\rangle$ and $|1\rangle$ correspond to the north and south poles, respectively.

The Bloch sphere provides an intuitive geometric description of the state of a single qubit. Multi-qubit systems require a Hilbert space of dimension 2^n , and entangled states cannot be represented by independent Bloch vectors. Apart from an overall global phase, every single-qubit unitary transformation can be interpreted as a rotation of the Bloch vector. This representation will be useful when introducing the Pauli operators and quantum gates.

1.3.2 Measurements and observables

The state of a quantum system cannot be accessed directly. Information about the system is obtained through a measurement process. When a qubit in the state $|\psi\rangle$ is measured in the computational basis $\{|0\rangle, |1\rangle\}$, the possible outcomes are 0 and 1. According to the Born rule, the probability of obtaining the outcome 0 is

$$P(0) = |\langle 0 | \psi \rangle|^2 = |\alpha|^2,$$

whereas the probability of obtaining the outcome 1 is

$$P(1) = |\langle 1 | \psi \rangle|^2 = |\beta|^2.$$

After an ideal projective measurement, the state is projected onto the state corresponding to the observed outcome. Therefore, if the outcome is 0, the post-measurement state is $|0\rangle$; if the outcome is 1, the post-measurement state is $|1\rangle$.

In quantum mechanics, a measurable physical quantity, or observable, is represented by a Hermitian operator $\hat{O}$ acting on the Hilbert space. Hermitian operators satisfy

$$\hat{O}^\dagger = \hat{O},$$

and consequently have real eigenvalues. The possible outcomes of an ideal measurement of an observable are its eigenvalues.

For a system prepared in the state $|\psi\rangle$, the expectation value of an observable $\hat{O}$ is given by

$$\langle \hat{O} \rangle = \langle \psi | \hat{O} | \psi \rangle.$$

This quantity represents the average measurement outcome obtained by repeating the same experiment many times on identically prepared systems. This concept will be particularly relevant in the following chapters, since QAOA evaluates an objective function through the expectation value of a cost Hamiltonian.

1.3.3 Time evolution and unitary transformations

In the absence of measurement and environmental interactions, the time evolution of a closed quantum system is governed by the Schrödinger equation,

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

where $\hat{H}$ is the Hamiltonian operator, which represents the total energy of the system.

For a time-independent Hamiltonian, the solution can be written as

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle,$$

where

$$\hat{U}(t) = e^{-i\hat{H}t/\hbar}$$

is the time-evolution operator. Since the Hamiltonian is Hermitian, the operator $\hat{U}(t)$ is unitary:

$$\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \mathbb{I}.$$

Unitary evolution preserves the normalization of quantum states and is reversible. In particular, for a unitary operator $\hat{U}$, the inverse transformation is given by its Hermitian conjugate,

$$\hat{U}^{-1} = \hat{U}^\dagger.$$

Therefore, in an ideal closed system, no quantum information is lost during the evolution, since the initial state can in principle be recovered by applying the inverse operation. This reversibility is a fundamental feature of coherent quantum dynamics.

1.4 Quantum gates and circuits

The circuit model translates coherent quantum dynamics into a computational framework. A quantum computation is described as a sequence of elementary unitary operations, called quantum gates, acting on one or more qubits. These operations transform the quantum state in a controlled way, while measurements performed at the end of the circuit produce classical outcomes and are not reversible.

A single-qubit gate is represented by a 2×2 unitary matrix. Among the most common examples are the Pauli gates

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and the Hadamard gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

On the Bloch sphere the Pauli operators correspond, up to a global phase, to rotations by an angle π about the x -, y -, and z -axes, respectively. In particular, the Pauli- X operator

exchanges the computational basis states,

$$\sigma_x|0\rangle = |1\rangle, \quad \sigma_x|1\rangle = |0\rangle,$$

and is therefore analogous to the classical NOT operation. More generally, rotations by an angle θ around the three Bloch-sphere axes are implemented by the gates

$$R_x(\theta) = e^{-i\theta\sigma_x/2}, \quad R_y(\theta) = e^{-i\theta\sigma_y/2}, \quad R_z(\theta) = e^{-i\theta\sigma_z/2}.$$

The Pauli matrices, together with the identity operator $\mathbb{I}$, form a basis for the vector space of 2×2 complex matrices. Therefore, any single-qubit operator $\hat{A}$ can be expressed as

$$\hat{A} = a_0\mathbb{I} + a_x\sigma_x + a_y\sigma_y + a_z\sigma_z,$$

where a_0 , a_x , a_y , and a_z are complex coefficients.

More specifically, apart from an overall global phase, every single-qubit unitary operation can be decomposed into a sequence of rotations about the Bloch-sphere axes. One possible decomposition is

$$\hat{U} = e^{i\delta}R_z(\alpha)R_y(\beta)R_z(\gamma),$$

where δ , α , β , and γ are real parameters. This result is particularly important when moving from the abstract circuit model to a physical quantum processor. A real quantum device cannot implement every possible unitary transformation directly. Instead, it provides a limited set of native gates, namely operations that can be physically realized and accurately calibrated on the chosen qubit platform.

Complex quantum circuits are therefore decomposed into sequences of native single-qubit rotations and entangling two-qubit gates. The specific native gate set depends on the physical implementation of the qubits and on the available control mechanisms. A compilation procedure, commonly referred to as transpilation, translates an abstract quantum circuit into an equivalent circuit that is compatible with the gate set, qubit connectivity, and operational constraints of a target device.

1.5 Physical implementations of qubits

The qubit introduced so far is an abstract mathematical model. To perform quantum computation, it must be encoded in a controllable physical system with at least two distinguishable quantum states, identified with $|0\rangle$ and $|1\rangle$. This can be achieved through several physical implementations, such as trapped ions, superconducting circuits, photons, or semiconductor spin systems. Regardless of the chosen platform, however, a physical qubit must satisfy certain properties in order to enable reliable and controllable quantum computation. These requirements are commonly summarized by the DiVincenzo criteria:

1. **Scalability:** the platform should support a large number of well-characterized qubits while keeping the required control resources and hardware overhead manageable.
2. **Initialization:** it must be possible to reliably prepare qubits in a known fiducial state, typically $|0\rangle^{\otimes n}$.
3. **Long coherence times:** quantum coherence should persist for a time much longer than the duration of elementary gate operations. In particular, the relevant relaxation and dephasing times should satisfy

$$T_1, T_2 \gg t_{\text{gate}}.$$

4. **Universal quantum gates:** the platform must provide a universal set of high-fidelity single- and two-qubit operations, allowing arbitrary quantum computations to be approximated to the desired accuracy.
5. **Qubit-specific measurement:** individual qubits must be measured with high accuracy in order to extract the result of the computation.

The requirements listed above describe the behavior of an ideal quantum processor. In practice, no existing hardware platform satisfies them perfectly. A physical qubit is an open quantum system: it interacts with the surrounding environment and is affected by imperfections in fabrication, calibration, control electronics, and measurement apparatus.

As a consequence, the real evolution of a quantum processor deviates from the ideal unitary circuit model. Quantum processors currently available are commonly referred to as Noisy Intermediate-Scale Quantum (NISQ) devices. The term, introduced by John Preskill in 2018, describes processors containing an intermediate number of physical qubits that are not protected by full quantum error correction. Their operations are therefore affected by decoherence, gate imperfections, and readout errors. Although NISQ processors can execute non-trivial quantum circuits, the accumulation of noise restricts the circuit depth and computational complexity that can be reliably achieved. In this work, the effects of readout errors, depolarizing noise, and thermal relaxation noise are investigated. These error mechanisms will be introduced in greater detail in the following chapters.

2. The Quantum Approximate Optimization Algorithm (QAOA)

The Quantum Approximate Optimization Algorithm (QAOA) is a hybrid quantum–classical variational algorithm designed to find approximate solutions to combinatorial optimization problems. Its operation is based on an iterative feedback loop between a parameterized quantum circuit and a classical optimizer. For a given set of variational parameters, the quantum circuit prepares a candidate quantum state and is repeatedly measured to estimate the value of a problem-dependent cost function. This estimate is then provided to a classical optimization routine, which updates the circuit parameters in order to improve the objective value. The updated parameters are used to execute the quantum circuit again, and the procedure is repeated until a predefined convergence criterion is satisfied.

The QAOA circuit is constructed by alternating two parameterized unitary operations: a cost unitary U_C , which encodes the objective function of the optimization problem, and a mixing unitary U_M , which drives the exploration of the space of candidate solutions. The algorithm was introduced by Farhi, Goldstone, and Gutmann in 2014.

2.1 Combinatorial optimization problems

Combinatorial optimization is a subfield of mathematical optimization concerned with finding an optimal solution within a finite set of possible configurations. The solution space is discrete, or can be reduced to a discrete representation.

Typical examples of combinatorial optimization problems include the traveling salesman problem, the minimum spanning tree problem, and the knapsack problem. For many relevant instances, the number of possible solutions grows exponentially with the problem size. As a consequence, exhaustive search rapidly becomes computationally impractical, and specialized exact methods, heuristics, or approximation algorithms are required to explore the search space efficiently.

2.2 The Max-Cut problem

In this work, QAOA is applied to the Max-Cut problem, one of the most widely studied problems in combinatorial optimization. Its relevance extends beyond graph theory, since Max-Cut provides a flexible mathematical framework for a broad range of discrete optimization tasks. These include graph partitioning, circuit layout design, scheduling, network optimization, clustering, and image analysis. Moreover, its formulation is closely related to several relevant problems in physics, including Ising models, spin-glass systems, and frustrated magnetic systems.

Applications of QAOA to physical systems will be discussed in Chapter 5. Let

$$G = (V, E)$$

be an undirected graph, where V is the set of vertices and E is the set of edges. A cut partitions the vertices into two disjoint subsets. An edge contributes to the cut if its two endpoints belong to different subsets. The Max-Cut problem consists of finding the partition that maximizes the total number of cut edges or, in the weighted case, the total weight of the cut edges.

A binary variable $x_i \in \{0, 1\}$ is assigned to each vertex $i \in V$, indicating to which subset of the partition the vertex belongs. For an unweighted graph, an edge (i, j) belongs to the cut when its endpoints are assigned different binary values, namely when $x_i \neq x_j$.

The contribution of a single edge to the cut value can be written as

$$C_{ij}(x) = x_i + x_j - 2x_i x_j. \quad (2.1)$$

Indeed, this expression is equal to 1 when $x_i \neq x_j$, and to 0 when $x_i = x_j$. The total cut value is therefore obtained by summing the contributions of all edges:

$$C(x) = \sum_{(i,j) \in E} (x_i + x_j - 2x_i x_j). \quad (2.2)$$

The optimal solution can be expressed as:

$$x^* = \arg \max_{x \in \{0,1\}^{|V|}} C(x).$$

x denotes a binary string of length $|V|$, where each component x_i specifies the subset to which the corresponding vertex belongs. The set $\{0, 1\}^{|V|}$ contains all possible binary assignments for the $|V|$ vertices of the graph and therefore represents the space of all possible bi-partitions.

The operator $\arg \max$ returns the value, or values, of the variable x for which the objective function $C(x)$ attains its largest value. Therefore, x^* represents an optimal binary assignment of the vertices, namely the partition that produces the maximum cut value. It is important to note that more than one optimal assignment may exist. In particular, exchanging the two subsets of a cut produces the complementary binary string and leaves the cut value unchanged. As will be shown in the chapter devoted to the numerical results, the specific graph instance considered in this work has a fourfold-degenerate optimal solution.

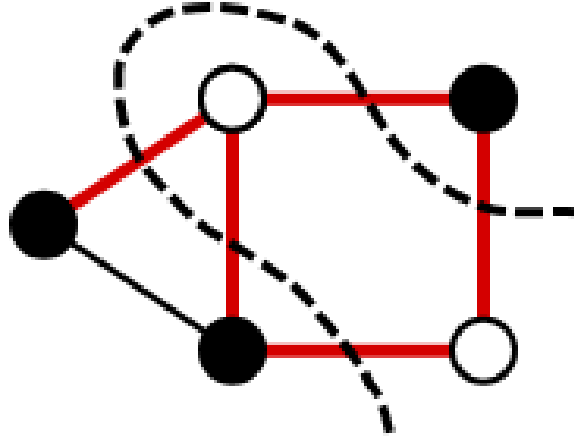


Figure 2.1: Example of a cut for a graph with five vertices. The black and white vertices represent the two subsets of the partition. Edges connecting vertices belonging to different subsets contribute to the cut value.

The binary formulation of Max-Cut provides a direct route to its quantum representation. By associating each binary variable with a qubit and replacing the classical variables with suitable quantum operators, the objective function can be encoded into a cost Hamiltonian. This Hamiltonian constitutes the problem-dependent component of the QAOA circuit.

2.3 QAOA formulation

The Quantum Approximate Optimization Algorithm (QAOA) is a hybrid quantum-classical variational algorithm designed to approximately solve combinatorial optimization problems. Its structure consists of an iterative feedback loop between a parameterized quantum circuit and a classical optimizer. The quantum circuit prepares a parameterized quantum state, while the classical optimizer updates the circuit parameters in order to improve the value of the objective function.

In the following section, the quantum-circuit framework underlying QAOA is introduced.

2.3.1 QAOA Quantum-circuit

The quantum circuit is based on the alternating application of two parameterized unitary operators: the cost operator $U_C(\gamma)$ and the mixer operator $U_M(\beta)$. These unitaries are generated by the cost Hamiltonian H_C and the mixer Hamiltonian H_M , respectively:

$$U_C(\gamma) = e^{-i\gamma H_C}, \quad U_M(\beta) = e^{-i\beta H_M}. \quad (2.3)$$

The cost Hamiltonian encodes the objective function of the optimization problem into a quantum operator. For the Max-Cut problem, each vertex is associated with a qubit and the classical binary variables are mapped onto Pauli-Z operators. For an unweighted graph $G = (V, E)$, the cost Hamiltonian can be written as

$$H_C = \frac{1}{2} \sum_{(i,j) \in E} (Z_i Z_j - \mathbb{I}). \quad (2.4)$$

For a computational-basis state $|x\rangle$, the eigenvalue associated with H_C is equal to the negative of the cut value represented by the corresponding bit string:

$$H_C |x\rangle = -C(x) |x\rangle. \quad (2.5)$$

In particular, an edge (i, j) contributes -1 when the two corresponding qubits have opposite values, and zero otherwise. Therefore, minimizing the expectation value of H_C is

equivalent to maximizing the expected Max-Cut value. The standard mixer Hamiltonian is defined as

$$H_M = \sum_{i \in V} X_i. \quad (2.6)$$

where X_i is the Pauli- X operator acting on the qubit associated with vertex i . Unlike the cost Hamiltonian, the mixer Hamiltonian is not diagonal in the computational basis. Its action induces transitions between different computational-basis states and therefore allows the algorithm to explore different candidate partitions. The mixer must not commute with the cost Hamiltonian; otherwise, its alternating application would not generate a non-trivial exploration of the solution space.

The parameters γ and β are variational angles associated with the cost and mixer evolutions, respectively. They can be interpreted as effective evolution times under H_C and H_M . Their values determine how strongly each operator modifies the quantum state: the cost parameter γ controls the phase separation induced by the objective function, whereas the mixer parameter β controls the amplitude transfer among different candidate solutions. These parameters are not fixed a priori; first they are randomly generated and then they are optimized by a classical routine in order to minimize the expectation value of the cost Hamiltonian.

The QAOA circuit is usually initialized in the uniform superposition

$$|\psi_0\rangle = |+\rangle^{\otimes n} = H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle,$$

where $n = |V|$ is the number of vertices of the graph. This state assigns the same initial probability amplitude to every possible binary assignment, and hence to every possible graph partition. It is obtained by applying a Hadamard gate to each qubit initially prepared in the state $|0\rangle$.

Starting from $|\psi_0\rangle$, the QAOA state at depth p is obtained by applying p alternating layers of cost and mixer unitaries:

$$|\psi(\gamma, \beta)\rangle = U_M(\beta_p) U_C(\gamma_p) \cdots U_M(\beta_1) U_C(\gamma_1) |\psi_0\rangle,$$

where

$$\gamma = (\gamma_1, \dots, \gamma_p), \quad \beta = (\beta_1, \dots, \beta_p).$$

The circuit therefore contains $2p$ variational parameters. Increasing the depth p increases the expressive power of the ansatz, since the algorithm can implement a more complex sequence of phase separation and mixing operations. However, it also increases the number of parameters to be optimized, the circuit depth, and the sensitivity to hardware noise.

Once the first evaluation of the quantum circuit has been completed, the resulting quantum state $|\psi(\gamma, \beta)\rangle$ is used to estimate the expectation value of the cost Hamiltonian H_C :

$$F_p(\gamma, \beta) = \langle \psi(\gamma, \beta) | H_C | \psi(\gamma, \beta) \rangle. \quad (2.7)$$

This quantity represents the expected Ising energy associated with the probability distribution generated by the QAOA circuit. With the Hamiltonian convention adopted in this work,

$$F_p(\gamma, \beta) = -\mathbb{E}[C]. \quad (2.8)$$

Therefore, minimizing F_p is equivalent to maximizing the expected Max-Cut value.

2.3.2 Classical Optimization Loop

The estimated value of $F_p(\gamma, \beta)$ is passed to a classical optimizer, which updates the variational parameters in order to minimize F_p and therefore increase the expected cut value.

Once the optimal parameters (γ^*, β^*) have been identified, the corresponding QAOA circuit is executed repeatedly and all qubits are measured in the computational basis. A single circuit execution, usually referred to as a shot, returns one bit string $x \in \{0, 1\}^n$, corresponding to a candidate partition of the graph.

The optimized quantum state can be expressed as

$$|\psi(\gamma^*, \beta^*)\rangle = \sum_{x \in \{0, 1\}^n} a_x |x\rangle. \quad (2.9)$$

where a_x is the probability amplitude associated with the computational basis state $|x\rangle$. According to the Born rule, measuring the circuit returns the bit string x with probability

$$P(x) = |a_x|^2 = |\langle x | \psi(\gamma^*, \beta^*) \rangle|^2. \quad (2.10)$$

Since each measurement collapses the quantum state to a single computational-basis state, one shot is not sufficient to characterize the output distribution. The circuit must therefore be executed for a finite number N_{shots} of shots. If a bit string x is observed n_x times, its probability is estimated by its relative frequency,

$$\hat{P}(x) = \frac{n_x}{N_{\text{shots}}}, \quad \sum_{x \in \{0,1\}^n} \hat{P}(x) = 1. \quad (2.11)$$

The resulting empirical distribution identifies the bit strings that are most likely to be measured for the optimized QAOA state. Each sampled bit string is then evaluated classically through the Max-Cut objective function $C(x)$. The final candidate solution can be selected as the most frequent bit string, or, more robustly, as the sampled bit string with the largest cut value.

2.4 Qiskit implementation

The numerical implementation was developed in Python using Qiskit, Qiskit Aer, and the rustworkx graph library. The algorithm was evaluated on an unweighted graph with five vertices and six edges,

$$E = \{(0,1), (0,2), (1,2), (0,4), (2,3), (3,4)\}.$$

Each vertex was mapped to one qubit and each edge was assigned unit weight.

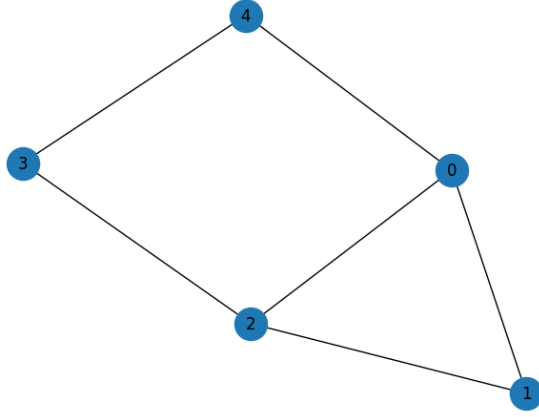


Figure 2.2: Unweighted graph used for the QAOA algorithm.

The maximum classical cut value for this graph is

$$C_{\max} = 5.$$

The optimal value is attained by four computational-basis bit strings, related by graph symmetries and by the global bit-flip transformation. For compatibility with the minimization interface of the classical optimizer, the Max-Cut problem was encoded through the Cost Hamiltonian

$$H_C = \frac{1}{2} \sum_{(i,j) \in E} (Z_i Z_j - \mathbb{I}). \quad (2.12)$$

For every computational-basis state $|x\rangle$, this operator satisfies

$$H_C |x\rangle = -C(x) |x\rangle. \quad (2.13)$$

where $C(x)$ is the corresponding classical cut value. Therefore, minimizing the expectation value of H_C is equivalent to maximizing the expected Max-Cut value:

$$\arg \min_{\theta} \langle \psi(\theta) | H_C | \psi(\theta) \rangle = \arg \max_{\theta} \langle C \rangle_{\theta}. \quad (2.14)$$

The Hamiltonian was represented using Qiskit's `SparsePauliOp` class. For each edge (i, j) , a Pauli string containing Z operators on qubits i and j , and identity operators on all other qubits, was added with coefficient $1/2$. The constant term $-|E|/2$ was represented through a single identity Pauli string.

The parameterized circuit was constructed using Qiskit's QAOAAnsatz class with depth

$$p = 3.$$

The ansatz consequently contains $2p = 6$ variational parameters. The circuit was compiled for the statevector-based AerSimulator backend using a preset pass manager with optimization level three. After transpilation, the cost Hamiltonian was transformed according to the layout of the compiled circuit in order to preserve a consistent qubit-to-operator mapping.

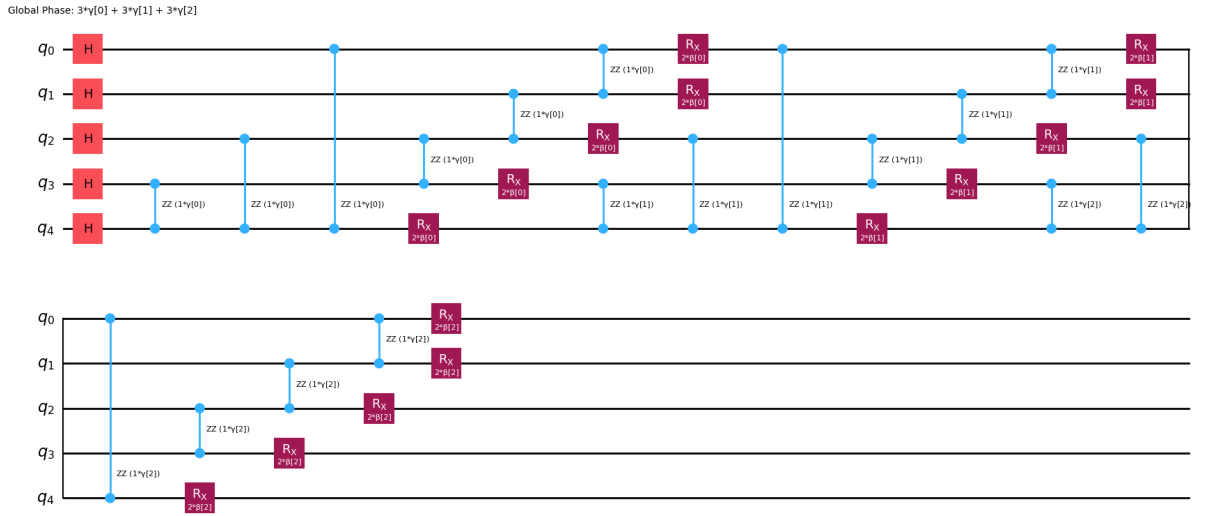


Figure 2.3: Parameterized QAOA circuit for the five-vertex Max-Cut instance at depth $p = 3$. The circuit contains an initial state-preparation stage and three alternating cost and mixer layers.

2.4.1 Objective-function evaluation

The numerical evaluation of the objective function was implemented through the `cost_func` routine. For a given vector of variational parameters θ , the routine passes the transpiled QAOA circuit, the corresponding cost Hamiltonian, and the numerical parameter values to Qiskit's EstimatorV2 primitive.

The quantity returned by `cost_func` is the expectation value

$$f(\theta) = \langle \psi(\theta) | H_C | \psi(\theta) \rangle. \quad (2.15)$$

This quantity represents the expected Ising energy of the parameterized QAOA state. Since the adopted Hamiltonian convention satisfies

$$H_C = -C,$$

minimizing $f(\theta)$ is equivalent to maximizing the expected Max-Cut value.

In practical terms, `cost_func` submits a primitive unified block (PUB) containing the parameterized circuit, the observable, and the current parameter vector. The expectation value returned by the estimator is then stored in a history array and passed to the classical parameter-update procedure. The stored values are subsequently used to analyze the convergence behaviour of the optimization process.

For the ideal reference simulations, the expectation value is evaluated exactly from the statevector. Consequently, the value returned by `cost_func` during the optimization is not affected by finite-shot statistical uncertainty.

The classical optimization strategies and the effect of finite-shot and hardware-induced noise on the objective-function evaluation are discussed in the following chapter.

Output bit-string distribution

After optimization, the final QAOA circuit is obtained by assigning the optimized parameter vector θ^* to the ansatz. In the ideal benchmark, the circuit is converted into a `Statevector` object using Qiskit's `Statevector.from_instruction` method. The probability distribution over computational-basis bit strings is then extracted directly through the `probabilities` method. Consequently, the ideal output distribution is evaluated exactly and is not affected by finite-shot statistical uncertainty.

For the noisy simulations, the optimized circuit is instead executed on the noisy simulator and measured repeatedly. The output distribution is obtained from the measurement counts and is therefore affected by finite-shot sampling and by the active noise model. The statistical properties of this estimation procedure are discussed in the following chapter.

3. Noise Models and Classical Optimizers

The QAOA formulation introduced in the previous chapter assumes that the expectation value of the cost Hamiltonian can be evaluated accurately for each set of variational parameters. In a realistic implementation, however, the value returned to the classical optimization procedure may be affected by finite sampling, imperfect quantum gates, decoherence, and measurement errors.

These effects can modify both the estimated expectation value of the cost Hamiltonian and the probability distribution over the measured bit strings. Consequently, they may alter the optimization landscape explored by the classical routine and reduce the probability of sampling high-quality solutions.

This chapter first introduces the noise sources considered in the numerical simulations. The classical optimization methods are then described in light of the properties of the resulting noisy objective function. Finally, the experimental protocol used to compare their performance is presented.

3.1 Noise Sources in QAOA

3.1.1 Finite-Shot Estimation

In the noisy simulations, the expectation value of the cost Hamiltonian is estimated from a finite number of circuit measurements. Each circuit execution, referred to as a shot, returns one computational-basis bit string. The resulting measurement counts provide an empirical estimate of the output probability distribution.

Because the number of shots is finite, this empirical distribution differs from the ideal distribution by statistical sampling error. Consequently, the cost-function value supplied to the classical optimizer fluctuates from one evaluation to another, even when the circuit parameters are kept fixed. Let

$$P_{\theta}(x) = |\langle x | \psi(\theta) \rangle|^2 \quad (3.1)$$

denote the ideal probability of measuring the bit string $x \in \{0, 1\}^n$ for a given parameter vector θ . If the circuit is executed N_{shots} times and the outcome x is observed n_x times, its empirical probability is

$$\hat{P}_{\theta}(x) = \frac{n_x}{N_{\text{shots}}}. \quad (3.2)$$

In the limit of an arbitrarily large number of shots, the empirical probability converges to the ideal probability:

$$\hat{P}_{\theta}(x) \xrightarrow{N_{\text{shots}} \rightarrow \infty} P_{\theta}(x). \quad (3.3)$$

For a finite number of shots, however, the observed count n_x is a random variable. Therefore, the empirical probability $\hat{P}_{\theta}(x)$ fluctuates around the ideal value $P_{\theta}(x)$.

The estimated Max-Cut expectation value can be obtained by averaging the classical cost values associated with the sampled bit strings:

$$\langle \widehat{C} \rangle_{\theta} = \sum_{x \in \{0,1\}^n} C(x) \hat{P}_{\theta}(x) = \frac{1}{N_{\text{shots}}} \sum_{s=1}^{N_{\text{shots}}} C(x^{(s)}), \quad (3.4)$$

where $x^{(s)}$ denotes the bit string obtained at shot s .

Using the Ising convention adopted in this work, the corresponding estimate of the cost Hamiltonian is

$$\hat{f}(\theta) = \langle \widehat{H_C} \rangle_{\theta} = -\langle \widehat{C} \rangle_{\theta}. \quad (3.5)$$

Consequently, the classical optimizer does not necessarily receive the exact value

$$f(\theta) = \langle \psi(\theta) | H_C | \psi(\theta) \rangle. \quad (3.6)$$

Instead, it may receive the finite-shot estimate $\hat{f}(\theta)$, which fluctuates from one evaluation to another.

The magnitude of the statistical uncertainty decreases as the number of shots increases. In general, the standard error of a sample mean scales as

$$\mathcal{O}\left(\frac{1}{\sqrt{N_{\text{shots}}}}\right). \quad (3.7)$$

Therefore, increasing N_{shots} produces a more accurate estimate of the output distribution and of the expectation value. However, it also increases the number of circuit executions required for each evaluation of the objective function.

Finite-shot uncertainty is present whenever the expectation value is estimated from a finite number of circuit measurements, even in the absence of gate and readout errors. In the ideal baseline considered in this work, both the expectation value of the cost Hamiltonian and the output bit-string probability distribution are extracted exactly from the statevector. Consequently, neither the optimization trajectory nor the reported output distribution is affected by finite-shot statistical uncertainty.

In the noisy simulations, the circuit is instead evaluated through a finite number of shots. The probability distribution over bit strings is therefore estimated from measurement counts and is affected by finite-shot sampling uncertainty, in addition to the active physical noise channels. The estimated cost-function value supplied to the classical optimizer consequently fluctuates between repeated evaluations at identical variational parameters.

3.1.2 Depolarizing Noise

Depolarizing noise is a quantum error model that describes the partial loss of information from a quantum state. Unlike finite-shot uncertainty, which arises from the statistical estimation of measurement probabilities, depolarizing noise acts directly on the quantum state during the execution of the circuit.

For an n -qubit density operator ρ , the depolarizing channel can be written as

$$\mathcal{E}_{\text{dep}}(\rho) = (1 - \lambda)\rho + \lambda \frac{\mathbb{I}}{2^n}, \quad (3.8)$$

where λ is the depolarizing parameter and $\mathbb{I}/2^n$ is the maximally mixed state. The

channel leaves the state unchanged when $\lambda = 0$, while for $\lambda = 1$ it maps the state to the maximally mixed state.

For a single qubit, the maximally mixed state is

$$\frac{\mathbb{I}}{2} = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|). \quad (3.9)$$

Consequently, as the depolarizing parameter increases, the qubit loses both population information and phase coherence. In the Bloch-sphere representation, the state vector is progressively contracted towards the center of the sphere.

The depolarizing channel can also be interpreted as a random Pauli error. For a single qubit, it can be written as

$$\mathcal{E}_{\text{dep}}(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z), \quad (3.10)$$

where p is the probability that a non-identity Pauli error occurs. This formulation makes explicit that the state may be affected by bit-flip, phase-flip, or combined bit-and-phase-flip errors.

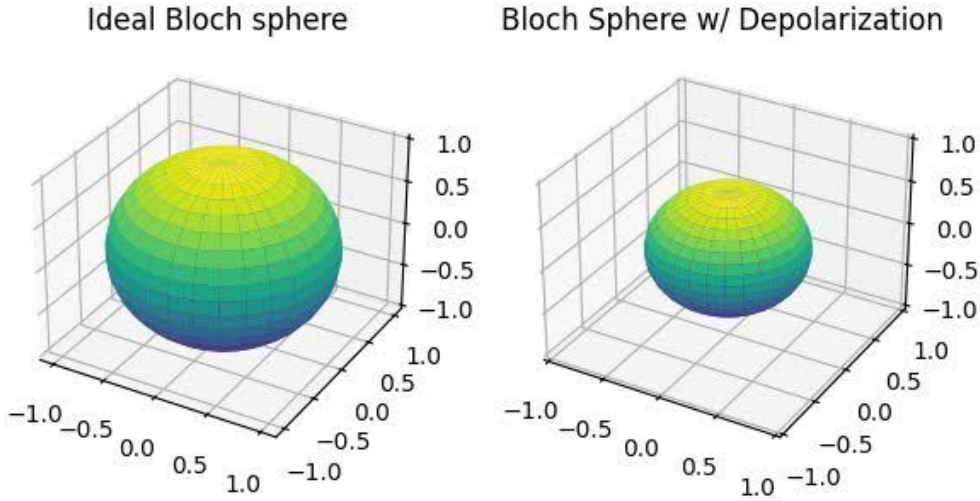


Figure 3.1: Geometrical representation of depolarizing noise on the Bloch sphere. The action of the depolarizing channel contracts the Bloch vector towards the center of the sphere, corresponding to the maximally mixed state.

3.1.3 Readout Error

Readout error, also referred to as measurement error, occurs during the final measurement of the quantum circuit. In this case, the quantum state may be correctly prepared immediately before the measurement, but the classical bit value recorded by the measurement apparatus does not necessarily coincide with the actual measurement outcome.

For a single qubit, readout error can be described by the assignment matrix

$$M = \begin{pmatrix} P(0|0) & P(0|1) \\ P(1|0) & P(1|1) \end{pmatrix}, \quad (3.11)$$

where $P(a|b)$ denotes the probability of recording the classical value a when the ideal measurement outcome is b . Each column of the matrix therefore sums to one:

$$P(0|b) + P(1|b) = 1, \quad b \in \{0, 1\}. \quad (3.12)$$

In the symmetric readout-error model, the probability of misidentifying a qubit prepared in $|0\rangle$ as $|1\rangle$ is equal to the probability of misidentifying $|1\rangle$ as $|0\rangle$. Denoting this probability by ϵ_{ro} , the assignment matrix becomes

$$M_{\text{sym}} = \begin{pmatrix} 1 - \epsilon_{\text{ro}} & \epsilon_{\text{ro}} \\ \epsilon_{\text{ro}} & 1 - \epsilon_{\text{ro}} \end{pmatrix}. \quad (3.13)$$

For example, if a circuit ideally produces the bit string 01001, a readout error can cause one or more recorded bits to be flipped. The observed outcome may therefore be a different bit string, such as 01011 or 11001, even though the ideal state immediately before measurement was unchanged.

Unlike depolarizing noise, readout error does not modify the coherent evolution generated by the cost and mixer unitaries. Instead, it transforms the ideal output probability distribution into a distorted measured distribution. If $P_{\text{ideal}}(x)$ is the probability of an ideal bit string x , the observed distribution can be expressed as

$$P_{\text{obs}}(y) = \sum_x P(y|x) P_{\text{ideal}}(x), \quad (3.14)$$

where $P(y|x)$ is the conditional probability of recording the bit string y when the true outcome is x .

In the Max-Cut formulation, every measured bit string corresponds to a graph partition. A readout error can therefore map a bit string representing a high-quality partition onto a different string associated with a lower cut value. This effect is particularly relevant when the final solution is selected from the observed bit-string distribution, because the most frequently recorded string is not necessarily the most probable string of the ideal QAOA state.

3.1.4 Thermal Relaxation

Thermal relaxation describes the loss of quantum information caused by the interaction between a qubit and its surrounding environment. In contrast to readout error, which acts only during the final measurement, thermal relaxation affects the quantum state throughout the execution of the circuit.

The model is characterized by two time constants: the energy-relaxation time T_1 and the coherence time T_2 . These quantities describe two different physical processes and determine how rapidly the state of a qubit deviates from its ideal evolution.

- The time constant T_1 , usually referred to as the energy-relaxation time, characterizes the exchange of energy between a qubit and its environment. A qubit in the excited state $|1\rangle$ can spontaneously decay towards the ground state $|0\rangle$, releasing energy into the environment.

For an initial excited-state population $P_1(0)$, the population remaining in $|1\rangle$ after an elapsed time t decreases approximately as

$$P_1(t) = P_1(0)e^{-t/T_1}. \quad (3.15)$$

Therefore, a long T_1 corresponds to slow energy relaxation and is desirable for quantum computation. When the duration of a gate or of an entire circuit is not negligible compared with T_1 , the probability of an unintended transition from $|1\rangle$ to $|0\rangle$ increases.

On the Bloch sphere, energy relaxation drives the state towards the equilibrium state, which at sufficiently low temperature is close to $|0\rangle$. Thus, the Bloch vector is progressively attracted towards the north pole of the sphere.

- The time constant T_2 , often called the coherence time or dephasing time, describes the loss of a well-defined relative phase between the components $|0\rangle$ and $|1\rangle$ of a superposition state.

Consider a qubit prepared in the state

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle). \quad (3.16)$$

The information encoded in this state is not only contained in the probabilities of measuring $|0\rangle$ and $|1\rangle$, which are both equal to $1/2$, but also in the fact that the two components have a definite relative phase. Environmental fluctuations can randomize this phase without necessarily changing the two measurement probabilities.

The effect is more clearly expressed using the density matrix:

$$\rho_+ = |+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \quad (3.17)$$

The diagonal terms describe the populations of $|0\rangle$ and $|1\rangle$, whereas the off-diagonal terms encode their phase coherence. Under dephasing, the off-diagonal terms decay approximately as

$$\rho_{01}(t) = \rho_{01}(0)e^{-t/T_2}. \quad (3.18)$$

As this coherence is lost, the state evolves from a coherent superposition towards a statistical mixture. In the Bloch-sphere representation, dephasing contracts the Bloch vector towards the vertical z -axis, without directly forcing it towards either pole.

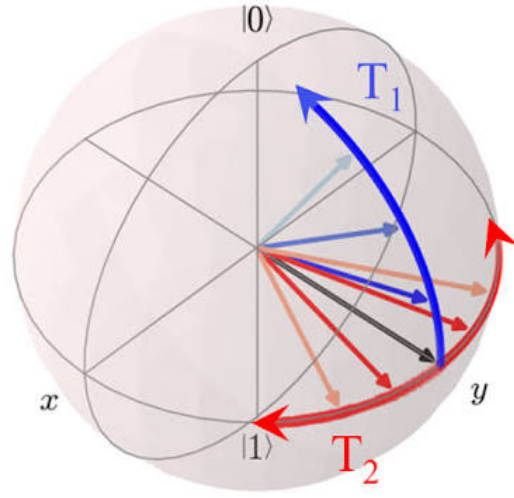


Figure 3.2: Schematic representation of thermal relaxation in a qubit. The time T_1 characterizes energy relaxation from the excited state $|1\rangle$ towards the ground state $|0\rangle$, while T_2 characterizes the loss of phase coherence in a superposition state.

In a physical qubit, energy relaxation also contributes to the loss of phase coherence. For this reason, the coherence time cannot be arbitrarily larger than the energy-relaxation time. The parameters must satisfy

$$T_2 \leq 2T_1. \quad (3.19)$$

The equality $T_2 = 2T_1$ corresponds to the limiting case in which decoherence arises only from energy relaxation. When

$$T_2 < 2T_1, \quad (3.20)$$

additional pure-dephasing mechanisms are present. These mechanisms randomize the relative phase of a superposition without requiring an energy exchange between the qubit and the environment.

3.2 Classical Optimizers

As discussed in the previous sections, finite sampling and physical noise can affect the expectation value of the cost Hamiltonian returned by the quantum circuit. The classical optimization procedure must therefore identify suitable QAOA parameters using an objective function that may differ from its ideal, noise-free counterpart.

The numerical analysis considers three optimization methods: COBYLA, SPSA, and L-BFGS-B. These methods were selected because they rely on different strategies to update the variational parameters. The following subsections describe the main features of the selected methods. Their performance is subsequently assessed for the QAOA Max-Cut instance under ideal and noisy simulation conditions.

3.2.1 COBYLA

COBYLA, an acronym for Constrained Optimization BY Linear Approximations, is a derivative-free optimization method. It is designed for problems in which the objective function can be evaluated at selected parameter values but its derivatives are not directly available.

At each step, COBYLA constructs a local linear approximation of the objective function from previously evaluated points and uses this model to propose a new parameter vector. The method can also handle inequality constraints, although no explicit parameter constraints are required in the present implementation.

For QAOA, COBYLA can optimize the variational parameters by using only the expectation values returned by the quantum circuit. It does not require an analytical gradient or a separate quantum-gradient evaluation. This makes it a convenient baseline method for variational quantum algorithms.

However, the quality of its local linear approximation may be reduced when the cost-function evaluations are strongly affected by sampling fluctuations or physical noise. Its performance in such conditions is therefore assessed numerically in the following chapters.

3.2.2 SPSA

Simultaneous Perturbation Stochastic Approximation (SPSA) is a stochastic optimization method designed for objective functions affected by statistical fluctuations. Rather than estimating each component of the gradient separately, SPSA perturbs all variational parameters simultaneously and constructs an approximate search direction from only two evaluations of the objective function.

Let $\theta^{(k)}$ denote the parameter vector at iteration k . SPSA generates a random perturbation vector

$$\Delta^{(k)} = \left(\Delta_1^{(k)}, \dots, \Delta_d^{(k)} \right), \quad (3.21)$$

whose components are commonly chosen independently from the set $\{-1, +1\}$. The objective function is then evaluated at two perturbed parameter vectors:

$$\theta^{(k)} + c_k \Delta^{(k)} \quad \text{and} \quad \theta^{(k)} - c_k \Delta^{(k)}, \quad (3.22)$$

where $c_k > 0$ determines the perturbation magnitude at iteration k .

An approximation of the i -th component of the gradient is obtained as

$$\hat{g}_i^{(k)} = \frac{f\left(\theta^{(k)} + c_k \Delta^{(k)}\right) - f\left(\theta^{(k)} - c_k \Delta^{(k)}\right)}{2c_k \Delta_i^{(k)}}. \quad (3.23)$$

The variational parameters are subsequently updated according to

$$\theta^{(k+1)} = \theta^{(k)} - a_k \hat{g}^{(k)}, \quad (3.24)$$

where $a_k > 0$ is the learning-rate parameter.

The main advantage of SPSA is that each iteration requires only two objective-function evaluations, independently of the number of variational parameters. In contrast, a conventional finite-difference gradient calculation requires multiple evaluations that increase with the dimension of the parameter space. This feature is particularly useful for variational quantum algorithms, in which each objective-function evaluation may require a large number of circuit executions.

Because SPSA uses random perturbations and is based on noisy function evaluations, the optimization trajectories can exhibit significant run-to-run variability. However, this stochastic structure can also make the method suitable for objective functions affected by finite-shot fluctuations and hardware noise.

3.2.3 L-BFGS-B

L-BFGS-B, short for Limited-memory Broyden–Fletcher–Goldfarb–Shanno with Bounds, is a quasi-Newton optimization method for the minimization of scalar functions. The method uses information from previous parameter updates and gradient estimates to construct an approximation of the local curvature of the objective function.

In a full Newton method, the search direction is obtained by using the Hessian matrix, which contains the second derivatives of the objective function. Computing and storing this matrix can become expensive as the number of parameters increases. L-BFGS-B avoids this issue by storing only a limited number of previous parameter displacements and gradient variations. This information is used to construct a limited-memory approximation of the inverse Hessian matrix.

At iteration k , the parameter and gradient variations can be written as

$$\mathbf{s}_k = \boldsymbol{\theta}^{(k+1)} - \boldsymbol{\theta}^{(k)}, \quad (3.25)$$

and

$$\mathbf{y}_k = \nabla f(\boldsymbol{\theta}^{(k+1)}) - \nabla f(\boldsymbol{\theta}^{(k)}). \quad (3.26)$$

The pairs $(\mathbf{s}_k, \mathbf{y}_k)$ from a limited number of previous iterations are used to approximate the inverse Hessian. The resulting search direction is then employed in a line-search procedure to determine the next parameter update.

The suffix “B” indicates that the method can enforce bound constraints on the optimization variables. In the present context, bounds can be used to restrict the search to a selected domain of QAOA angles, although the implementation may also be executed without explicit bounds.

L-BFGS-B is particularly effective when the objective function is smooth and can be evaluated accurately. In the ideal statevector simulations considered in this work, the expectation value of the cost Hamiltonian is computed exactly, so the method can exploit reliable local information to search for a minimum efficiently.

In the presence of finite-shot fluctuations or physical noise, numerical gradient estimates and curvature information can become less reliable. As a result, the local model constructed by L-BFGS-B may be distorted, and its performance can become more sensitive to the noise level than that of methods specifically designed for stochastic optimization.

3.2.4 Expected Behaviour Under Noise

The three optimization methods use different types of information to update the QAOA parameters. COBYLA relies on local linear approximations without explicit derivatives, SPSA estimates a stochastic search direction from simultaneous parameter perturbations, and L-BFGS-B exploits local gradient and curvature information.

These differences suggest that the methods may respond differently when the objective function is affected by finite-shot uncertainty or physical noise. SPSA is expected to be naturally suited to stochastic objective functions, whereas L-BFGS-B may benefit from the exact and smooth objective landscape available in the ideal statevector simulation. COBYLA provides a derivative-free reference method whose performance depends on the reliability of its local linear approximation.

These considerations provide qualitative expectations only. The numerical experiments presented in the following chapter determine the actual performance of the three optimizers for the selected Max-Cut instance and noise configurations.

3.3 Experimental Protocol

All numerical simulations were implemented in Python using Qiskit, Qiskit Aer, and Qiskit Algorithms. Qiskit was used to construct the QAOA circuit and the cost Hamiltonian, while Qiskit Aer was used to perform ideal and noisy simulations. The classical

optimization routines were imported from the `qiskit_algorithms.optimizers` module.

Noisy simulations were implemented by constructing a Qiskit Aer NoiseModel and passing it to an AerSimulator backend. Depolarizing noise was generated using the `depolarizing_error` function, with separate parameters controlling the error strength of one-qubit and two-qubit gates. Readout error was defined through the `ReadoutError` class, whose assignment matrix specifies the probability that a measured classical bit is recorded incorrectly. Thermal relaxation was implemented with the thermal relaxation error function, using the parameters T_1 , T_2 , and the duration associated with the affected gate.

The optimization procedures were implemented using the `COBYLA`, `SPSA`, and `L-BFGS-B` classes. Each method received the same QAOA objective function, namely the expectation value of the cost Hamiltonian. The parameter `maxiter` defines the maximum number of iterations allowed in an optimization run. Depending on the selected method, the procedure may stop earlier when its internal convergence condition is satisfied. Additional optimizer-specific hyperparameters control the update procedure and the stopping conditions. For example, `COBYLA` uses parameters related to the size and termination tolerance of its local search region. `SPSA` is controlled by the learning-rate and perturbation schedules, whereas `L-BFGS-B` uses settings related to numerical-gradient estimation, convergence tolerances, and the maximum number of function evaluations.

For all simulations, the same graph instance, QAOA depth, cost Hamiltonian, and overall workflow were used. The numerical values assigned to the noise parameters and optimizer settings, together with the resulting performance measures, are reported in the following chapter.

4. Numerical Experiments and Results

This chapter presents the numerical results obtained by applying QAOA to the Max-Cut problem introduced in the previous chapters. The analysis compares three classical optimizers, namely COBYLA, SPSA, and L-BFGS-B, first in an ideal setting and then under different quantum-noise models. The objective is to assess not only the final solution quality obtained by each optimizer, but also its stability with respect to random parameter initialization and its sensitivity to different sources and strengths of noise.

Ideal-state benchmark. The analysis begins with an ideal statevector benchmark, in which neither quantum noise nor finite-shot sampling is included. This reference configuration provides an exact and deterministic cost function and therefore represents the most favourable setting for the optimizers. It makes it possible to characterize their intrinsic behaviour before introducing the statistical fluctuations associated with finite-shot sampling and noisy quantum operations.

Exploratory noise analysis. The noisy analysis is organized in two stages. First, an exploratory study is performed for each individual noise channel: depolarizing noise, readout error, and thermal relaxation. For every noise channel, three parameter regimes, referred to as low, medium, and high noise, are considered. Each configuration is evaluated over 20 independent runs to identify the qualitative effect of increasing noise strength and to compare the behavior of the three optimizers.

The selected parameter ranges are synthetic and are not intended to reproduce the calibration data of a single quantum processor exactly. They were instead selected as synthetic values with orders of magnitude informed by calibration data from contemporary superconducting quantum processors. For example, an IBM Quantum calibration snapshot for `ibm_marrakesh` reports median two-qubit and readout errors of 2.881×10^{-3} and 1.453×10^{-2} , respectively, together with median coherence times of $T_1 = 171.91 \mu\text{s}$ and $T_2 = 66.5 \mu\text{s}$ [7]. The values used in this section are not intended to reproduce this device exactly; instead, they define controlled low-, medium-, and high-noise regimes for

studying the relative behavior of the optimizers.

Large-scale validation. The exploratory study is followed by a large-scale validation based on 100 independent runs for one selected configuration of each noise channel. The configuration selected for the extended analysis was determined by the magnitude of the performance degradation observed in the exploratory study. The aim was to focus on noise regimes that produce a measurable effect on the QAOA optimization while avoiding cases in which the noise is either too weak to reveal meaningful differences or so strong that the optimization performance is severely suppressed.

For depolarizing noise, the medium-noise configuration was selected because it produced an intermediate but clearly observable degradation with respect to the ideal benchmark. For readout error and thermal relaxation, the low-noise configurations were selected because they already produced a substantial performance degradation. The 100-run simulations therefore provide a more reliable statistical characterization of the optimizer performance in the noise regimes where the effect of the corresponding error source is most informative.

Combined noise model. Finally, a combined low-noise model is considered, in which depolarizing noise, readout error, and thermal relaxation are simultaneously included. This model is designed as a controlled synthetic scenario in which the three main error mechanisms of a noisy quantum computation act together. Its parameters do not exactly coincide with the low-noise configurations used in the individual-channel studies; rather, they were selected to remain within realistic order-of-magnitude ranges while maintaining a regime in which optimization is still feasible. In this context, the term “low-noise” therefore denotes the relative severity of the combined model within this study, rather than an exact calibration of a specific hardware device.

Common experimental setup. All simulations use the five-vertex Max-Cut graph introduced previously and a QAOA ansatz with depth $p = 3$. For each run, the variational parameters are initialized randomly in the interval $[0, 2\pi]$. The parameter `maxiter` does not have an identical interpretation for all optimizers. For COBYLA, it specifies the maxi-

maximum number of objective-function evaluations. For SPSA, it specifies the maximum number of parameter-update iterations; since the stochastic-gradient estimate requires multiple objective-function evaluations per iteration, the total number of evaluations can be larger than `maxiter`. For L-BFGS-B, `maxiter` limits the number of quasi-Newton iterations. Consequently, it should be interpreted as an optimizer-specific computational budget rather than as a common and directly comparable number of quantum-circuit executions. The ideal benchmark is evaluated using exact statevector expectation values. In the noisy experiments, the cost function is estimated from measured bit-string counts using a finite number of shots. Unless otherwise stated, all noisy simulations use 4096 shots for each cost-function evaluation.

For every configuration, the reported standard deviation is the sample standard deviation across independent optimization runs. It quantifies the run-to-run variability arising from different random initial parameters and from stochastic cost-function evaluations caused by finite-shot sampling and noise. It does not represent a separately propagated shot uncertainty within a single final cost-function evaluation.

The cost Hamiltonian is defined as

$$H_C = \frac{1}{2} \sum_{(i,j) \in E} (Z_i Z_j - I), \quad (4.1)$$

so that

$$\langle H_C \rangle = -\mathbb{E}[C]. \quad (4.2)$$

Therefore, more negative values of $\langle H_C \rangle$ correspond to larger expected Max-Cut values.

Interpretation of convergence and output distributions. To provide a more complete view of the optimization process, the results are reported not only through final objective-value statistics, but also through mean convergence curves and output bit-string distributions. The convergence curves show the evolution of the cost-function value $\langle H_C \rangle$ as a function of the number of objective-function evaluations. Each point corresponds to one evaluation of the cost function, that is, to the execution of a parameterized QAOA circuit and the subsequent estimation of $\langle H_C \rangle$.

The horizontal axis therefore does not necessarily correspond one-to-one to the iteration counter of the classical optimizer. Depending on the optimization method and on its internal convergence procedure, multiple cost-function evaluations can be performed within a single optimization iteration. The curves are obtained by recording the value returned by every cost-function evaluation during each run, aligning the trajectories to their common available length, and then computing the mean and standard deviation across the independent runs.

The output bit-string distributions are evaluated only after the optimization has terminated. For each run, the optimized parameters returned by the classical optimizer are assigned to the QAOA circuit, the circuit is measured, and it is sampled using the specified number of shots. The resulting counts are normalized by the total number of shots to estimate the probability of each computational-basis bit string. When a mean distribution is reported, the estimated bit-string probabilities are averaged over the independent runs. These distributions make it possible to assess whether the optimized circuit concentrates probability on bit strings associated with high-quality, and in particular optimal, Max-Cut solutions.

4.1 Ideal-State Benchmark

As stated previously, this section establishes a reference for the best-case performance of the three classical optimizers. The QAOA circuit is evaluated in an ideal setting, in which neither quantum noise nor finite-shot sampling is included. Consequently, the observed differences between optimizers arise only from their optimization strategies and from the choice of the initial variational parameters.

The ideal simulations are performed using a statevector representation of the quantum state. Starting from the initial state $|0\rangle^{\otimes n}$, the statevector simulator first applies Hadamard gates to all qubits to create the uniform superposition state $|+\rangle^{\otimes n}$, then applies all gates of the parameterized QAOA circuit and computes the complete final state

$$|\psi(\theta)\rangle = \sum_x c_x(\theta) |x\rangle, \quad (4.3)$$

where the complex amplitudes $c_x(\theta)$ fully determine the probability distribution of all computational-basis bit strings,

$$P(x) = |c_x(\theta)|^2. \quad (4.4)$$

Rather than executing repeated measurements and estimating the cost from a finite set of sampled bit strings, the expectation value of the cost Hamiltonian is computed directly as

$$\langle H_C \rangle = \langle \psi(\theta) | H_C | \psi(\theta) \rangle. \quad (4.5)$$

Therefore, no shots are required and no sampling uncertainty is introduced. For unitary circuits, the statevector estimator returns an exact expectation value, apart from the negligible numerical precision of the classical simulation. This produces a deterministic and smooth objective function, which provides the most favourable optimization landscape for the three classical methods.

Table 4.1: Final objective values obtained by COBYLA, SPSA, and L-BFGS-B in the ideal-state benchmark over 100 independent runs.

Optimizer	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$	Minimum	Median
COBYLA	−4.5983	0.3194	−4.8262	−4.6986
SPSA	−4.3879	0.3822	−4.8106	−4.5066
L-BFGS-B	−4.7420	0.1507	−4.8262	−4.7540

L-BFGS-B achieved the lowest mean final value of the cost Hamiltonian, $\langle H_C \rangle = -4.7420$, together with the smallest standard deviation, $\sigma = 0.1507$. This indicates that, in the ideal setting, L-BFGS-B provides both the best average optimization performance and the highest stability across different initial parameter vectors. COBYLA obtained an intermediate mean value of -4.5983 with a larger standard deviation of 0.3194 , whereas SPSA reached a mean value of -4.3879 and showed the largest variability, with a standard deviation of 0.3822 .

These results are consistent with the properties of the three optimization methods. In the ideal setting, the objective function is smooth and deterministic, allowing L-BFGS-B

to exploit local gradient information effectively. SPSA, although particularly suitable for noisy objective functions, does not offer the same advantage when the function can be evaluated exactly.

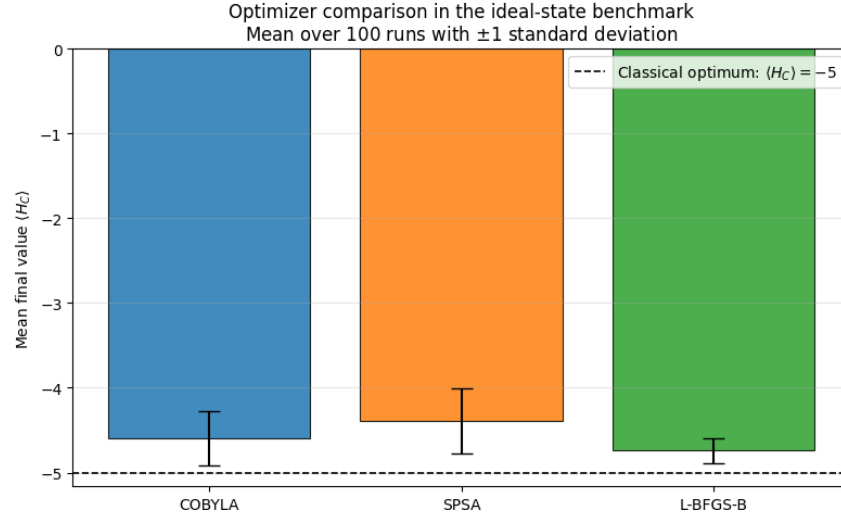
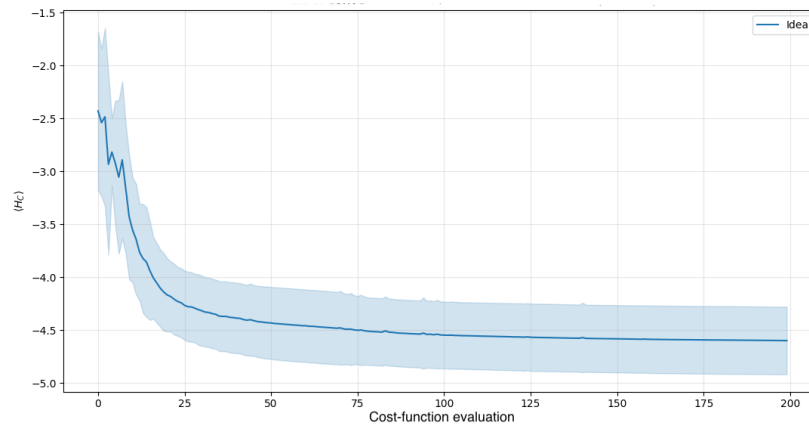


Figure 4.1: Mean final objective values obtained by COBYLA, SPSA, and L-BFGS-B in the ideal-state benchmark over 100 independent runs. The dashed line indicates the exact classical optimum, $\langle H_C \rangle = -5$, corresponding to a maximum cut value of 5.



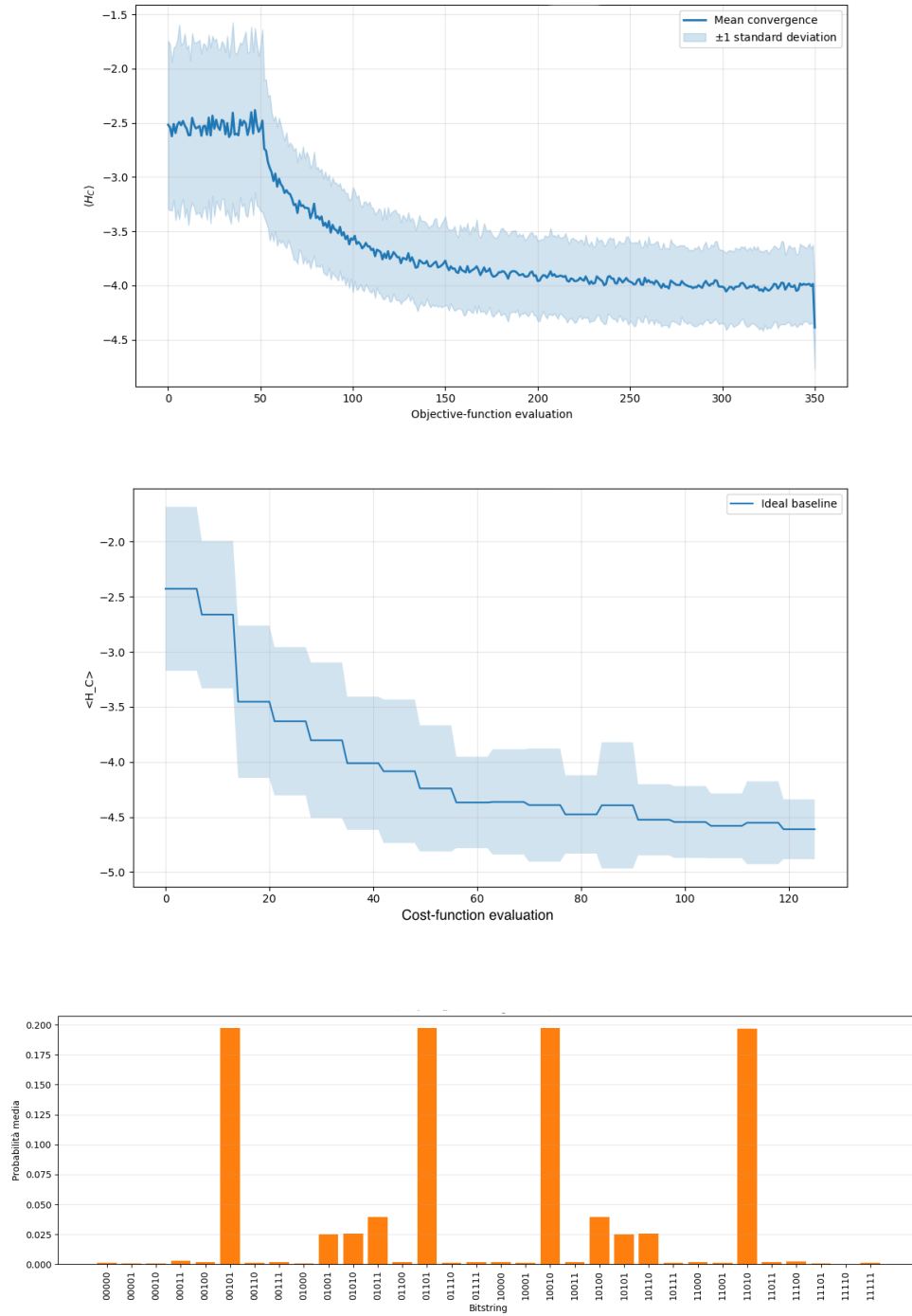


Figure 4.2: Mean output bit-string distribution obtained with L-BFGS-B in the ideal-state benchmark.

The mean bit-string distribution confirms the fourfold degeneracy of the Max-Cut problem discussed in the previous chapters. Four bit strings corresponding to optimal

cuts receive the largest and approximately equal probabilities, each close to 0.20. This reflects the fact that the optimal solutions are related by the symmetries of the graph and by the global bit-flip transformation, and therefore have the same cut value. The remaining probability is distributed among suboptimal configurations.

4.2 Depolarizing Noise

Having established the ideal-state benchmark, the effect of depolarizing noise is investigated. In these simulations, depolarizing channels were associated with the one- and two-qubit gates of the transpiled QAOA circuit. Unlike the ideal-state benchmark, the objective function was estimated from finite-shot measurement outcomes. Therefore, the following results include both depolarizing noise and shot noise.

The analysis was performed following the structure described in the introduction.

Table 4.2: Depolarizing-noise configurations used in the exploratory study. The parameters p_{1q} and p_{2q} denote the depolarizing probabilities applied to one-qubit and two-qubit gates, respectively.

Noise level	p_{1q}	p_{2q}
Low	0.001	0.005
Medium	0.003	0.010
High	0.005	0.020

4.2.1 Exploratory Study: 20 Runs

COBYLA

Table 4.3: COBYLA performance under depolarizing noise in the exploratory 20-run study.

Level	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$	Minimum	Maximum	Median
Low	-4.5420	0.2770	-4.7965	-3.6056	-4.6408
Medium	-4.5341	0.2849	-4.7631	-3.4601	-4.6188
High	-4.5112	0.2845	-4.7525	-3.4701	-4.5981

SPSA

Table 4.4: SPSA performance under depolarizing noise in the exploratory 20-run study.

Level	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$	Minimum	Maximum	Median
Low	-4.3560	0.4366	-4.7876	-3.4973	-4.5513
Medium	-4.4910	0.3362	-4.7824	-3.2661	-4.5757
High	-4.3766	0.3682	-4.7586	-3.4806	-4.5430

L-BFGS-B

Table 4.5: L-BFGS-B performance under depolarizing noise in the exploratory 20-run study.

Level	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$	Minimum	Maximum	Median
Low	-4.5260	0.1135	-4.6210	-4.0939	-4.5468
Medium	-4.2587	0.2768	-4.4193	-3.4356	-4.3436
High	-4.0289	0.1646	-4.1257	-3.3661	-4.0641

Comparison

Table 4.3–4.5 show that, when using the same shot budget, number of iterations, ansatz ISA and estimator-based cost function, COBYLA, SPSA, and L-BFGS-B achieve comparable mean final values of $\langle H_C \rangle$ under depolarizing noise. For the low-noise configuration, COBYLA and L-BFGS-B obtain very similar means (-4.5420 and -4.5260 , respectively), while SPSA is slightly less negative (-4.3560) but with a larger standard deviation (0.4366 vs. 0.2770 – 0.1135). In the medium- and high-noise regimes, the differences between optimizers remain small compared to the run-to-run variability. In particular, COBYLA exhibits the most stable behavior across noise levels, with means ranging from -4.5420 (low) to -4.5112 (high) and standard deviations around 0.28 . L-BFGS-B shows a more pronounced degradation as the noise increases, with the mean worsening from -4.5260 (low) to -4.0289 (high), but retains the smallest variability in the low-noise case. SPSA displays intermediate mean performance and the largest run-to-run fluctuations, especially at low and medium noise levels. Overall, no optimizer exhibits a clear systematic advantage over the others across all depolarizing configurations; instead, the choice of optimizer mainly affects the trade-off between mean performance and stability.

4.2.2 Large-Scale Validation: 100 Runs

To assess the robustness of the preliminary findings, we repeated the depolarizing-noise experiment with a larger number of independent runs. We focused on the medium-noise configuration used in the exploratory study, namely $p_{1q} = 0.003$ and $p_{2q} = 0.010$, and we increased the number of runs from 20 to 100 while keeping all other settings unchanged ($p = 3$, shots = 4096, maxiter = 150). Table 4.6 summarizes the mean, standard deviation, minimum, maximum, and median of the final $\langle H_C \rangle$ for the optimizers.

Table 4.6: Performance of COBYLA, SPSA, and L-BFGS-B under medium depolarizing noise ($p_{1q} = 0.003$, $p_{2q} = 0.010$) over 100 independent runs.

Optimizer	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$	Minimum	Maximum	Median
COBYLA	-4.5559	0.2906	-4.7902	-3.3379	-4.6465
SPSA	-4.3846	0.3412	-4.7757	-3.4856	-4.5093
L-BFGS-B	-4.3222	0.1503	-4.4193	-3.4356	-4.3436

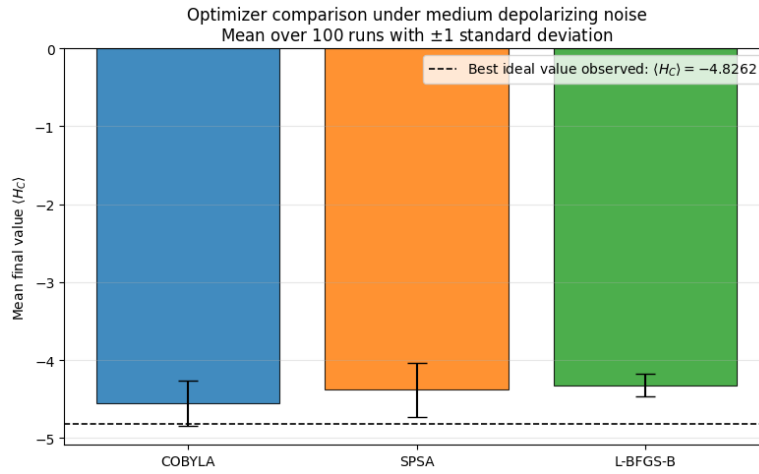


Figure 4.3: Comparison of the mean final cost-Hamiltonian values obtained by COBYLA, SPSA, and L-BFGS-B under medium depolarizing noise over 100 independent runs. The dashed horizontal line denotes the best final value observed in the ideal-state benchmark, $\langle H_C \rangle = -4.8262$.

Comparison

Table 4.6 shows that, in the medium-noise regime and with a larger number of runs, COBYLA achieves the most negative mean final value of $\langle H_C \rangle$ (-4.5559), followed by SPSA (-4.3846) and L-BFGS-B (-4.3222). COBYLA also exhibits relatively large variability ($\sigma = 0.2906$), intermediate between SPSA ($\sigma = 0.3412$) and L-BFGS-B ($\sigma = 0.1503$). These results confirm that, under depolarizing noise, all three optimizers achieve comparable mean performance, with differences smaller than the run-to-run fluctuations.

COBYLA tends to obtain slightly better mean values than SPSSA and L-BFGS-B in this setting, while L-BFGS-B displays the smallest variability across runs. SPSSA shows intermediate mean performance and the largest spread of final values.

4.3 Readout Error

4.3.1 Exploratory Study: Readout Error

We repeat the analysis of the previous section for the readout error. We consider three noise levels, characterized by the probabilities $p_{0|1}$ and $p_{1|0}$ of flipping the outcomes $|1\rangle \rightarrow |0\rangle$ and $|0\rangle \rightarrow |1\rangle$, respectively (Table 4.7). All experiments use $p = 3$, $\text{maxiter} = 150$, $\text{shots} = 4096$ and 20 independent runs.

Table 4.7: Readout-error configurations used in the exploratory study. The parameters $p_{0|1}$ and $p_{1|0}$ denote the probabilities of flipping $|1\rangle \rightarrow |0\rangle$ and $|0\rangle \rightarrow |1\rangle$, respectively.

Noise level	$p_{0 1}$	$p_{1 0}$
Low	0.01	0.02
Medium	0.03	0.04
High	0.06	0.06

COBYLA

Table 4.8: COBYLA performance under readout noise in the exploratory 20-run study.

Level	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$	Minimum	Maximum	Median
Low	-3.8955	0.1526	-4.3545	-3.6279	-3.8765
Medium	-3.8356	0.1599	-4.1924	-3.4951	-3.8267
High	-3.7230	0.1170	-3.9209	-3.4160	-3.7349

SPSA

Table 4.9: SPSA performance under readout noise in the exploratory 20-run study.

Level	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$	Minimum	Maximum	Median
Low	−4.0530	0.1835	−4.3975	−3.6865	−4.0415
Medium	−3.8903	0.2037	−4.3018	−3.5654	−3.8853
High	−3.8239	0.1465	−4.1543	−3.5801	−3.7974

L-BFGS-B

Table 4.10: L-BFGS-B performance under readout noise in the exploratory 20-run study.

Level	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$	Minimum	Maximum	Median
Low	−2.5431	0.4896	−3.2607	−1.5254	−2.7866
Medium	−2.5817	0.4754	−3.2705	−1.4863	−2.8237
High	−2.6184	0.4193	−3.1826	−1.7607	−2.7798

Comparison

Tables 4.8–4.10 show that, under readout noise, SPSA and COBYLA achieve substantially more negative mean final values of $\langle H_C \rangle$ than L-BFGS-B across all noise levels. For the low-noise configuration, SPSA attains the best mean performance (−4.0530), followed by COBYLA (−3.8955) and L-BFGS-B (−2.5431). This ranking is preserved at medium and high noise levels, with SPSA and COBYLA remaining close to each other and L-BFGS-B consistently worse.

In terms of stability, COBYLA exhibits the smallest standard deviations ($\sigma \approx 0.12$ – 0.16), while SPSA shows slightly larger fluctuations ($\sigma \approx 0.15$ – 0.20) and L-BFGS-B displays the largest variability ($\sigma \approx 0.42$ – 0.49). These results suggest that, in the presence of readout errors, gradient-free methods such as COBYLA and SPSA are more robust than

the gradient-based L-BFGS-B, at least for this Max-Cut instance and the chosen parameter settings.

4.3.2 Large-Scale Validation: 100 Runs

As done for depolarizing noise, we assess the robustness of the preliminary findings by increasing the number of independent runs. We focus on the low readout-noise configuration ($p_{0|1} = 0.01$, $p_{1|0} = 0.02$) and increase the number of runs from 20 to 100, keeping all other settings unchanged ($p = 3$, $\text{maxiter} = 150$, $\text{shots} = 4096$). Table 4.11 summarizes the mean, standard deviation, minimum, maximum, and median of the final $\langle H_C \rangle$.

Table 4.11: Performance of COBYLA, SPSA, and L-BFGS-B under low readout noise ($p_{0|1} = 0.01$, $p_{1|0} = 0.02$) over 100 independent runs.

Optimizer	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$	Minimum	Maximum	Median
COBYLA	-3.9156	0.1701	-4.4502	-3.5381	-3.9355
SPSA	-4.0574	0.2075	-4.5400	-3.7402	-4.0239
L-BFGS-B	-2.4739	0.6118	-3.6709	-0.6250	-2.5000

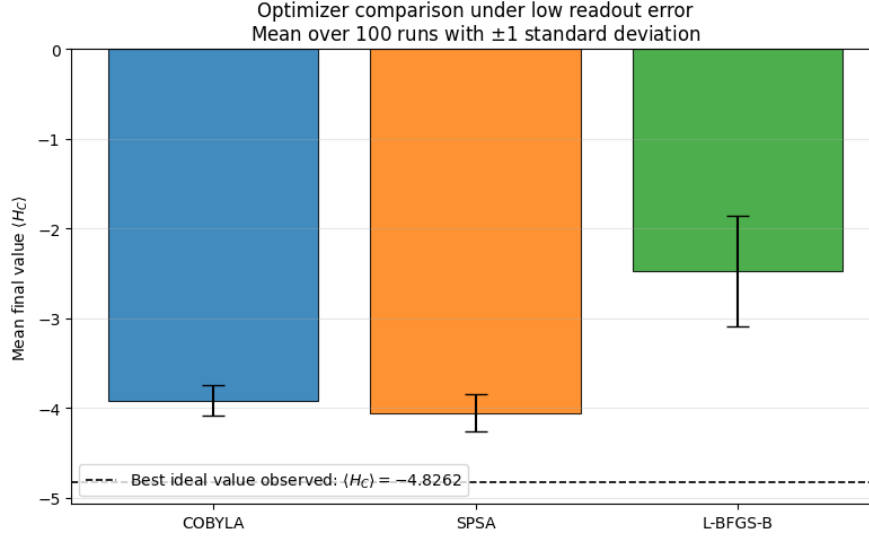


Figure 4.4: Comparison of the mean final cost-Hamiltonian values obtained by COBYLA, SPSA, and L-BFGS-B under low readout error over 100 independent runs.

Comparison

Table 4.11 shows that, under low readout noise and with 100 independent runs, SPSA achieves the most negative mean final value of $\langle H_C \rangle$ (-4.0574), followed by COBYLA (-3.9156) and L-BFGS-B (-2.4739). SPSA and COBYLA exhibit comparable performance, with SPSA slightly better on average but with larger variability ($\sigma = 0.2075$ vs. $\sigma = 0.1701$). L-BFGS-B displays substantially worse mean performance and the largest spread of final values ($\sigma = 0.6118$), confirming the trend observed in the exploratory 20-run study. These results indicate that, in the presence of readout errors, gradient-free optimizers (SPSA and COBYLA) are more effective and stable than the gradient-based L-BFGS-B for this Max-Cut instance.

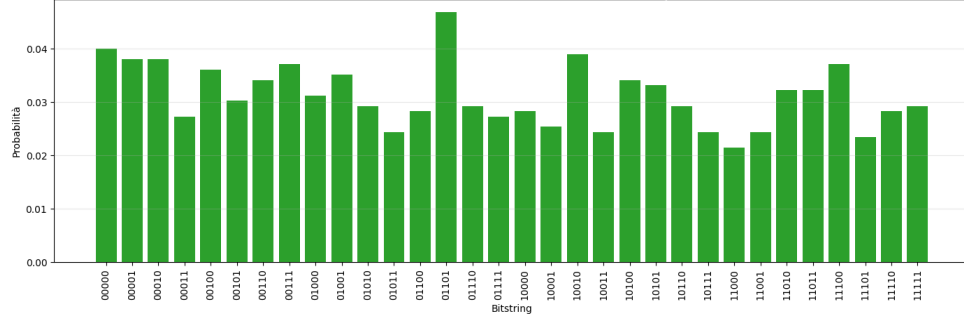


Figure 4.5: Mean output bit-string distribution obtained with L-BFGS-B under low read-out error.

The Bit-string distribution confirms the results, showing no optimal-cut string prevailing among the others. This is the easiest way to understand how noise influences the measures.

4.4 Thermal Relaxation Noise

4.4.1 Exploratory Study: Thermal Relaxation

Next, we investigate the impact of thermal relaxation errors on QAOA performance. We consider three noise levels, characterized by the relaxation times T_1 and T_2 (in nanoseconds), and fixed gate durations of $t_{1q} = 50\text{ns}$ for single-qubit gates and $t_{2q} = 300\text{ns}$ for two-qubit gates. All experiments use QAOA depth $p = 3$, shots = 4096 and maxiter = 200 for the final measurement. Maxiter was increased from 150 to 200 because the corresponding computational time remained limited. Each configuration is evaluated over 20 independent runs.

The thermal-relaxation configurations are summarized in Table 4.12.

Table 4.12: Thermal-relaxation configurations used in the exploratory study. The parameters T_1 and T_2 denote the relaxation and dephasing times (in ns), respectively.

Noise level	T_1 (ns)	T_2 (ns)
Low	200 000	150 000
Medium	80 000	60 000
High	30 000	20 000

COBYLA

Table 4.13: COBYLA performance under thermal relaxation noise in the exploratory 20-run study.

Level	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$
Low	−3.9755	0.1640
Medium	−3.8652	0.1478
High	−3.6371	0.1586

SPSA

Table 4.14: SPSA performance under thermal relaxation noise in the exploratory 20-run study.

Level	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$
Low	−4.0941	0.1961
Medium	−4.0338	0.1860
High	−3.7413	0.1327

L-BFGS-B

Table 4.15: L-BFGS-B performance under thermal relaxation noise in the exploratory 20-run study.

Level	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$
Low	−2.4469	0.5123
Medium	−2.4708	0.4858
High	−2.5808	0.3662

Comparison

Tables 4.13–4.15 show that, under thermal relaxation noise, SPSA and COBYLA achieve substantially more negative mean final values of $\langle H_C \rangle$ than L-BFGS-B across all noise levels. For the low-noise configuration, SPSA attains the best mean performance (−4.0941), followed by COBYLA (−3.9755) and L-BFGS-B (−2.4469). This ranking is preserved at medium and high noise levels, with SPSA and COBYLA remaining close to each other and L-BFGS-B consistently worse.

In terms of stability, COBYLA exhibits the smallest standard deviations ($\sigma \approx 0.14$ – 0.19), while SPSA shows slightly larger fluctuations ($\sigma \approx 0.13$ – 0.20) and L-BFGS-B displays the largest variability ($\sigma \approx 0.37$ – 0.51). These results suggest that, in the presence of thermal relaxation errors, gradient-free methods such as COBYLA and SPSA are more robust than the gradient-based L-BFGS-B, at least for this Max-Cut instance and the chosen parameter settings.

4.4.2 Large-Scale Validation: 100 Runs

As done for depolarizing and readout noise, we validate the robustness of the preliminary findings by increasing the number of independent runs. We focus on the low configuration of thermal-relaxation used in the exploratory study ($T_1 = 200000$ ns, $T_2 = 150000$ ns) and increase the number of runs from 20 to 100, keeping all other settings unchanged

($p = 3$, $\text{maxiter} = 200$, $\text{shots} = 4096$). Table 4.16 reports the corresponding statistics for COBYLA, SPSA, and L-BFGS-B.

Table 4.16: Performance of COBYLA, SPSA, and L-BFGS-B under thermal-relaxation low noise ($T_1 = 200\,000$ ns, $T_2 = 150\,000$ ns) over 100 independent runs.

Optimizer	$\overline{\langle H_C \rangle}$	$\sigma_{\langle H_C \rangle}$	Minimum	Maximum	Median
COBYLA	-3.9570	0.1496	-4.4565	-3.5981	-3.9824
SPSA	-4.1191	0.2132	-4.6929	-3.6680	-4.0823
L-BFGS-B	-2.4669	0.6435	-3.5869	-0.5454	-2.5149

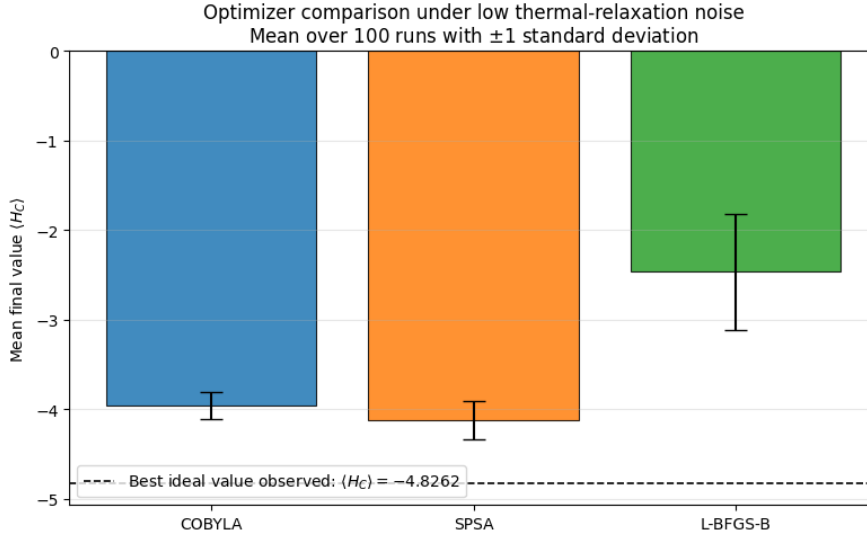


Figure 4.6: Comparison of the mean final cost-Hamiltonian values obtained by COBYLA, SPSA, and L-BFGS-B under low thermal-relaxation noise over 100 independent runs.

Comparison

Table 4.16 shows that, under thermal-relaxation low noise and with 100 independent runs, SPSA achieves the most negative mean final value of $\langle H_C \rangle$ (-4.1191), followed by COBYLA (-3.9570) and L-BFGS-B (-2.4669). SPSA and COBYLA exhibit comparable performance, with SPSA slightly better on average but with larger variability

($\sigma = 0.2132$ vs. $\sigma = 0.1496$). L-BFGS-B displays substantially worse mean performance and the largest spread of final values ($\sigma = 0.6435$), confirming the trend observed in the exploratory 20-run study. These results indicate that, in the presence of thermal relaxation errors, gradient-free methods (SPSA and COBYLA) are more effective and stable than the gradient-based L-BFGS-B for this Max-Cut instance.

4.5 Combined Low-Noise Model

In the previous sections, the effects of depolarizing noise, readout errors, and thermal relaxation were analyzed separately. However, in a real quantum computer, these noise mechanisms act simultaneously: gates are subject to coherent and incoherent errors, measurements are affected by readout errors, and qubits decay toward the ground state during circuit execution. To obtain a more realistic model, this section considers a combined low-noise model in which all three noise channels are active simultaneously.

The noise parameters were chosen to represent a device with performance comparable to current mid-range superconducting hardware. Specifically:

- **Depolarizing noise:** error probabilities $p_{1q} = 0.001$ for single-qubit gates and $p_{2q} = 0.01$ for two-qubit gates, in line with typical CX gate error rates on real devices.
- **Readout error:** the assignment matrix

$$M = \begin{pmatrix} 0.98 & 0.01 \\ 0.02 & 0.99 \end{pmatrix}, \quad (4.6)$$

corresponding to a probability of $P(1|0) = 0.02$ of recording 1 when the ideal outcome is 0, and a probability of $P(0|1) = 0.01$ of recording 0 when the ideal outcome is 1.

- **Thermal relaxation:** coherence times $T_1 = 300 \mu s$ and $T_2 = 250 \mu s$, with gate durations of 50ns (1q) and 300ns (2q), representative of well-isolated superconducting qubits.

The “low” noise level was chosen because it represents a regime where the noise is sufficiently low to allow the optimizer to converge, but high enough to significantly degrade performance compared to the ideal case. This scenario is therefore representative of execution on real hardware with accurate calibration.

For each optimizer (COBYLA, SPSA, L-BFGS-B), 100 independent runs were performed with QAOA at depth $p = 3$ and 4096 shots per cost function evaluation. The final statistics are reported in the following tables. In addition to the final expectation value, the probability of sampling an optimal solution was evaluated. Let $\mathcal{S}_{\text{opt}}$ denote the set of bit strings attaining the classical maximum cut value $C_{\text{max}} = 5$. The corresponding probability is defined as

$$P(\text{opt}) = \sum_{x \in \mathcal{S}_{\text{opt}}} P(x). \quad (4.7)$$

For each optimizer, the reported value is the mean of $P(\text{opt})$ over the 100 independent optimization runs.

4.5.1 COBYLA

Table 4.17: Final statistics for COBYLA in the combined low-noise model (100 runs, $p = 3$, 4096 shots).

Mean $\langle H_C \rangle$	Std $\langle H_C \rangle$	Mean $\mathbb{E}[C]$	Mean $P(\text{opt})$	Mean time/run	Total time
-3.747	0.120	3.747	0.236	0.14 min	0.23 h

COBYLA shows good optimization capability even in the presence of all active noise channels. The mean value of $\langle H_C \rangle = -3.75$ corresponds to an expected cut of approximately 3.75, close to the theoretical optimum for the considered graph. The contained standard deviation (0.12) indicates good stability across runs. The mean probability of sampling an optimal solution is $P(\text{opt}) = 0.236$.

Table 4.18: Final statistics for SPSA in the combined low-noise model (100 runs, $p = 3$, 4096 shots).

Mean $\langle H_C \rangle$	Std $\langle H_C \rangle$	Mean $\mathbb{E}[C]$	Mean $P(\text{opt})$	Mean time/run	Total time
-3.876	0.146	3.876	0.332	0.61 min	1.01 h

4.5.2 SPSA

SPSA achieves the best mean value of $\langle H_C \rangle = -3.876$, corresponding to an expected cut of 3.876, slightly higher than that obtained by COBYLA. The standard deviation of 0.146 indicates good robustness across the independent runs. The mean probability of sampling an optimal solution is $P(\text{opt}) = 0.332$, which is the highest among the three optimizers. The computational cost is higher than for COBYLA, with a mean time per run of approximately 0.61 minutes and a total execution time of 1.01 hours for the 100 runs.

4.5.3 L-BFGS-B

Table 4.19: Final statistics for L-BFGS-B in the combined low-noise model (100 runs, $p = 3$, 4096 shots).

Mean $\langle H_C \rangle$	Std $\langle H_C \rangle$	Mean $\mathbb{E}[C]$	Mean $P(\text{opt})$	Mean time/run	Total time
-2.594	0.493	2.594	0.088	0.31 min	0.51 h

L-BFGS-B shows significantly inferior performance compared to the other two optimizers. The mean value of $\langle H_C \rangle = -2.59$ corresponds to an expected cut of only 2.60, far from the optimum. The high standard deviation (0.49) indicates strong variability across runs, with some cases converging to poor-quality local minima. The probability of optimal solution is only 8.8%.

4.5.4 Comparison

Table 4.20: Comparison of the three optimizers in the combined low-noise model.

Metric	COBYLA	SPSA	L-BFGS-B
Mean $\langle H_C \rangle$	-3.747	-3.876	-2.594
Std $\langle H_C \rangle$	0.120	0.146	0.493
Mean $\mathbb{E}[C]$	3.747	3.876	2.594
Mean $P(\text{opt})$	0.236	0.332	0.088
Mean time/run	0.14 min	0.61 min	0.31 min

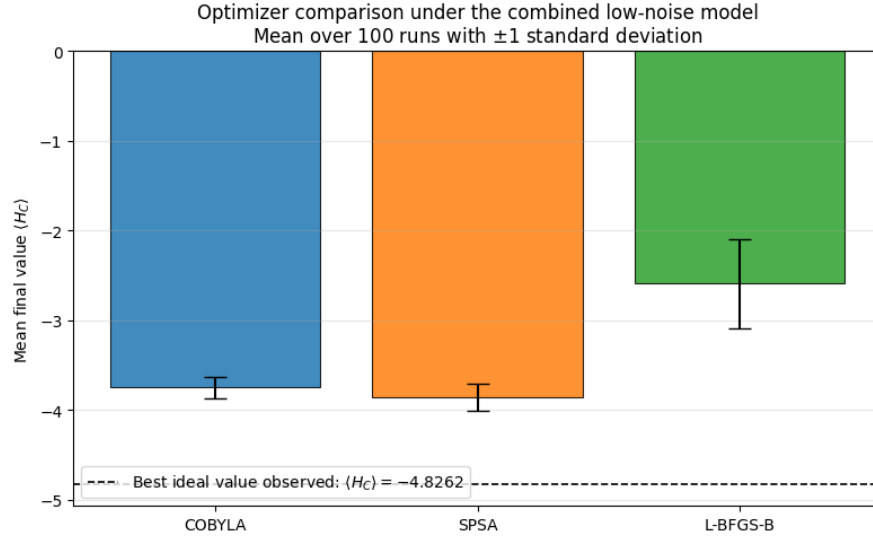


Figure 4.7: Comparison of the mean final cost-Hamiltonian values obtained by COBYLA, SPSA, and L-BFGS-B under the combined low-noise model over 100 independent runs.

The results show that:

- **Solution quality:** SPSA achieves the best mean final value of the cost Hamiltonian, $\langle H_C \rangle = -3.876$, corresponding to an expected cut of 3.876. COBYLA follows with $\langle H_C \rangle = -3.747$ and an expected cut of 3.747, whereas L-BFGS-B reaches only 2.594.
- **Probability of optimal solutions:** SPSA also has the highest mean probability of sampling an optimal cut, $P(\text{opt}) = 0.332$, followed by COBYLA with 0.236 and L-BFGS-B with 0.088.
- **Stability:** COBYLA shows the lowest standard deviation, 0.120, followed by SPSA, 0.146. L-BFGS-B is much more variable, with a standard deviation of 0.493, indicating greater sensitivity to random initializations and a tendency to converge to lower-quality solutions.
- **Computational efficiency:** COBYLA is the fastest, with a mean execution time of 0.14 minutes per run, followed by L-BFGS-B with 0.31 minutes and SPSA with 0.61 minutes per run.

In summary, in the combined low-noise model **SPSA and COBYLA are significantly superior to L-BFGS-B** in terms of both mean solution quality and stability. COBYLA offers the best trade-off between quality, stability, and computational efficiency, while SPSA achieves the best mean quality but at a higher computational cost. L-BFGS-B, despite its strong performance in the ideal-state benchmark, shows substantially worse results in the combined low-noise model. This behavior is consistent with its reliance on numerical gradient estimates and local curvature information. In the presence of finite-shot fluctuations and physical noise, these local estimates can become unreliable, leading to inefficient parameter updates and increased sensitivity to random initializations.

5. Applications of QAOA to Physical and Real-World Problems

The Quantum Approximate Optimization Algorithm (QAOA) has emerged as one of the most promising variational quantum algorithms for tackling combinatorial optimization problems on near-term quantum devices. While QAOA was initially motivated by theoretical computer science, its natural formulation in terms of spin Hamiltonians makes it particularly well-suited for addressing problems arising in condensed matter physics, statistical mechanics, and beyond.

This chapter explores the range of problems that can be addressed using QAOA, from fundamental physical models such as the Ising model, spin glasses, and frustrated magnetic systems, to real-world optimization tasks in logistics, finance, and machine learning. The discussion begins with the Ising model and spin glasses as paradigmatic examples of hard optimization problems and connects them to the Max-Cut formulation studied in this work. It then considers geometrically frustrated spin systems, whose ground-state search also leads to structured Ising-like optimization problems. Finally, it examines practical QUBO-based applications that illustrate the potential role of QAOA in industrial and scientific optimization.

5.1 Spin Glasses and the Ising Model

5.1.1 The Ising Model as a Prototype of Max-Cut

The Ising model is one of the most studied models in statistical physics and serves as a canonical framework for understanding phase transitions, collective behavior, and emergent phenomena in magnetic systems. In its simplest form, the model consists of binary spin variables $s_i \in \{-1, +1\}$ arranged on a lattice, with an energy function (Hamiltonian)

given by:

$$H = - \sum_{i < j} J_{ij} s_i s_j - \sum_i h_i s_i, \quad (5.1)$$

where J_{ij} represents the interaction strength between spins i and j , and h_i denotes an external magnetic field acting on spin i . When all couplings J_{ij} are positive and uniform, the system favors alignment of neighboring spins, leading to ferromagnetic order at low temperatures. In contrast, when $J_{ij} < 0$, the system is antiferromagnetic and favors anti-alignment.

From a computational perspective, finding the ground state configuration of the Ising Hamiltonian, that is, the spin assignment that minimizes H , is equivalent to solving a quadratic unconstrained binary optimization (QUBO) problem. By mapping $s_i \in \{-1, +1\}$ to binary variables $x_i \in \{0, 1\}$ via $s_i = 2x_i - 1$, the Ising model can be rewritten in QUBO form:

$$\min_{x \in \{0,1\}^N} x^\top Q x + c^\top x, \quad (5.2)$$

where Q and c are derived from the couplings J_{ij} and fields h_i . This equivalence establishes the Ising model as a universal representation of NP-hard combinatorial optimization problems, making it a natural target for QAOA.

The Max-Cut problem studied in this work is a prominent instance of this framework. For the special case of zero external fields ($h_i = 0$) and uniform antiferromagnetic couplings ($J_{ij} < 0$ for connected vertices), the cost Hamiltonian takes the form:

$$H_C = \frac{1}{2} \sum_{(i,j) \in E} (Z_i Z_j - I), \quad (5.3)$$

where Z_i denotes the Pauli-Z operator acting on qubit i . Minimizing $\langle H_C \rangle$ corresponds to maximizing the expected cut value, which is precisely the objective that QAOA attempts to achieve through variational optimization of the parameterized quantum circuit.

5.1.2 Spin Glasses: Disorder and Frustration

Spin glasses extend the Ising model by introducing disorder and frustration into the interaction network. In a spin glass, the couplings J_{ij} are not uniform but are instead drawn

from a probability distribution, often taking both positive and negative values. This randomness leads to *frustration*: situations where no spin configuration can simultaneously satisfy all pairwise interactions. For example, consider three spins s_1, s_2, s_3 with couplings $J_{12} > 0$, $J_{23} > 0$, and $J_{13} < 0$. The first two couplings favor alignment ($s_1 = s_2$ and $s_2 = s_3$), while the third favors anti-alignment ($s_1 \neq s_3$), creating an unavoidable conflict.

Frustration results in a highly complex energy landscape with an exponential number of local minima, separated by large energy barriers. This makes finding the global minimum extremely challenging for classical optimization algorithms, which often become trapped in suboptimal configurations. Spin glasses thus represent a prototypical class of hard optimization problems that are ideal benchmarks for testing the performance of QAOA and other quantum optimization methods.

5.2 Frustrated Magnetism

5.2.1 Geometric Frustration in Magnetic Lattices

Frustrated magnetism arises when the geometry of a lattice prevents simultaneous minimization of all pairwise spin interactions, even in the absence of disorder. A classic example is the antiferromagnetic Ising model on a triangular lattice: each triangle contains three spins with antiferromagnetic couplings ($J < 0$), but it is impossible to arrange three spins such that all three pairs are anti-aligned. This *geometric frustration* leads to a massively degenerate ground state manifold and exotic phases of matter, such as spin liquids, where spins remain disordered even at zero temperature.

Other frustrated lattices include the kagome, pyrochlore, and honeycomb structures, which host rich physics including topological order, fractionalized excitations, and emergent gauge fields. The Hamiltonian for these systems often takes the Ising or Heisenberg form:

$$H = J \sum_{\langle i,j \rangle} s_i s_j \quad (\text{Ising}), \quad H = J \sum_{\langle i,j \rangle} \vec{S}_i \cdot \vec{S}_j \quad (\text{Heisenberg}), \quad (5.4)$$

where $\langle i, j \rangle$ denotes nearest-neighbor pairs.

5.2.2 QAOA for Frustrated Spin Systems

Frustrated magnetic systems present a natural application for QAOA, as their ground state search is equivalent to minimizing an Ising-like Hamiltonian with complex interaction patterns. Unlike spin glasses, where frustration arises from random couplings, frustrated magnets exhibit structured frustration determined by lattice geometry. This makes them ideal for studying how QAOA performs on problems with known symmetries and degeneracies.

Potential applications include:

- **Material Design:** Predicting magnetic order and phase diagrams for novel compounds with frustrated lattices.
- **Quantum Simulation:** Emulating frustrated spin models on quantum hardware to explore phenomena inaccessible to classical simulations.
- **Benchmarking:** Using frustrated magnets as testbeds to compare QAOA performance against classical heuristics like simulated annealing or parallel tempering.

5.3 Real-World Optimization Problems

5.3.1 Combinatorial Optimization in Industry

The equivalence between the Ising model and QUBO enables QAOA to address a wide range of practical optimization problems. Key examples include:

- **Max-Cut and Graph Partitioning:** Given a graph, partition vertices into two sets to maximize the weight of edges between sets. Applications include community detection in social networks, image segmentation, and circuit design.
- **Scheduling and Resource Allocation:** Assigning tasks to time slots, workers to shifts, or gates to flights while minimizing conflicts and costs.
- **Vehicle Routing and Logistics:** Optimizing delivery routes, supply chain networks, and transportation schedules to reduce fuel consumption and delivery times.

Each of these problems can be formulated as a QUBO and mapped to an Ising Hamiltonian, making them amenable to QAOA.

5.3.2 Machine Learning and Artificial Intelligence

Spin glass models also appear in machine learning, particularly in energy-based models and neural networks:

- **Hopfield Networks:** Associative memory models where stored patterns correspond to local minima of an Ising-like energy function.
- **Boltzmann Machines:** Probabilistic graphical models used for unsupervised learning, with energy functions identical to spin glass Hamiltonians.
- **Clustering and Classification:** Formulating data partitioning as a minimization problem over binary variables.

QAOA-inspired approaches may provide an alternative strategy for exploring low-energy configurations in such models; however, a practical advantage over classical methods remains an open question.

5.3.3 Finance and Risk Management

In finance, spin glass and Ising models are used to study systemic risk, market stability, and optimal investment strategies:

- **Systemic Risk:** Modeling interbank networks as spin systems, where the ground state represents a stable financial configuration.
- **Asset Selection:** Formulating portfolio construction as a QUBO problem to identify optimal subsets of assets.
- **Market Dynamics:** Simulating agent-based models where traders' decisions are influenced by interactions analogous to spin couplings.

For instance, the portfolio optimization problem can be written as:

$$\min_{x \in \{0,1\}^N} \lambda \sum_{i,j} \sigma_{ij} x_i x_j - \sum_i \mu_i x_i, \quad (5.5)$$

where σ_{ij} is the covariance between assets i and j , μ_i is the expected return, and λ controls the risk-return trade-off.

Conclusions

The aim of this work was to investigate the impact of finite-shot sampling and different quantum noise channels on the performance of the Quantum Approximate Optimization Algorithm (QAOA) applied to a Max-Cut problem. The QAOA circuit was studied through ideal and noisy simulations, including depolarizing noise, readout errors, thermal relaxation, and a combined-noise model.

The comparison among COBYLA, SPSA, and L-BFGS-B showed that the performance of the classical optimization procedure depends strongly on the statistical properties of the objective function, which are significantly shaped by the noise channels. In the ideal statevector benchmark, where the expectation value is evaluated exactly, L-BFGS-B achieved the best overall performance. When finite-shot sampling and physical noise were introduced, however, its performance became substantially more sensitive to fluctuations of the cost function, particularly under readout error and thermal relaxation. In the noisy regimes considered, SPSA generally provided the highest mean solution quality, whereas COBYLA offered a more favorable compromise between solution quality, stability across independent runs, and computational cost.

The pronounced degradation of L-BFGS-B under readout error and thermal relaxation, compared with its comparatively milder degradation under depolarizing noise, suggests that these two channels distort the local information used by the optimizer more severely. L-BFGS-B relies on numerical gradient estimates and on an approximation of the local curvature of the objective function. When the variation of the estimated cost function produced by a finite parameter displacement is comparable to the fluctuations induced by finite-shot sampling and physical noise, the finite-difference gradient estimate can be dominated by noise rather than by the true local slope. The resulting gradient and curvature information may therefore become unreliable, leading to inefficient parameter updates and larger run-to-run variability.

This interpretation is consistent with the observed results but was not directly verified in the present work. A systematic diagnosis would require evaluating the variance of

the cost-function estimator at fixed parameter values and comparing it with the objective variation generated by the finite-difference step used by L-BFGS-B. Such an analysis, together with a study of the dependence on the shot budget and gradient-estimation step size, is left for future work.

As expected, QAOA performance is significantly affected by finite-shot sampling and physical noise channels. Consequently, the choice of the classical optimizer is critical. The present study is limited to a small Max-Cut instance and to controlled synthetic noise models, which do not reproduce the complete behavior of a specific quantum processor. Nevertheless, it provides a systematic framework for analyzing the combined role of finite-shot uncertainty, gate noise, readout errors, and thermal relaxation in QAOA optimization.

Possible extensions of this work include the execution of the same QAOA configurations on real quantum hardware, the study of larger graph instances and higher circuit depths, and the optimization of both shot budgets and classical-optimizer parameters. The use of error-mitigation methods and, in the longer term, quantum error-correction techniques will be necessary to reduce the impact of physical errors, improve the reliability of the estimated cost function, and enable the transition from current NISQ devices toward fault-tolerant quantum computing.

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